# Variation in Narrative Identity Is Associated With Trajectories of Mental Health Over Several Years

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This article presents 2 longitudinal studies designed to assess the relationship between variability in narrative identity and trajectories of mental health over several years. In Study 1, core scenes from 89 late-mid-life adults' life stories were assessed for several narrative themes. Participants' mental health and physical health were assessed concurrently with the narratives and annually for the subsequent 4 years. Concurrent analyses indicated that the themes of agency, redemption, and contamination were significantly associated with mental health. Longitudinal analyses indicated that these same 3 themes were significantly associated with participants' trajectories of mental health over the course of 4 years. Exploratory analyses indicated that narratives of challenging experiences may be central to this pattern of results. In Study 2, similar longitudinal analyses were conducted on a sample of 27 late-mid-life adults who received a major physical illness diagnosis between the baseline assessment and 6 months later and a matched sample of 27 control participants who remained healthy throughout the study. Participants' mental health and physical health were assessed every 6 months for 2 years. In this study, the themes of agency, communion, redemption, and contamination in participants' life narratives collected at baseline (before any participant became sick) were significantly associated with mental health in the group of participants who went on to receive a medical diagnosis, but not in the control group. Taken together, the results of these 2 studies indicate that the way an individual constructs personal narratives may impact his or her trajectory of mental health over time.

*Keywords:* narrative identity, health, agency, communion, redemption, contamination

Theories of narrative identity suggest that adults make sense of their lives by constructing and telling stories about their experiences (e.g., Habermas & Bluck, 2000; Hammack, 2008; McAdams, 1995, 2001; McAdams & McLean, 2013; McAdams & Pals, 2006; McLean, Pasupathi, & Pals, 2007; Pasupathi, 2001; Sarbin, 1986; Singer, 2004). These narratives, which connect the reconstructed past to the perceived present and the anticipated future, serve to provide the self with a sense of purpose and unity (e.g., McAdams, 1995, 2001). Stories that thematically and structurally distill an especially strong sense of purpose and unity ought to be psychologically beneficial for their narrator, promoting health and well-being (e.g., Adler, 2012; Bauer & McAdams, 2010; Lodi-Smith, Geise, Roberts, & Robins, 2009; McAdams, 1995, 2001; McAdams & McLean, 2013). Without a doubt, purpose and unity are important ends in their own right and narrative identity is concerned with much more than just mental health (MH). For example, McAdams (2006) asserts that narrative identity should also provide convincing causal explanations for the self, reflect the richness of lived experience, and advance socially valued actions. Nonetheless, there is a strong expectation running through much of the research on narrative identity that "good stories" are those that support the narrator's MH (e.g., Adler, Lodi-Smith, Philippe, & Houle, in press).

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Researchers have empirically tested the relationship between narrative identity and MH in a number of different studies. Often, they have assessed narrative identity and MH in the wake of difficult life experiences, for hardships not only threaten MH but also pose a challenge to narrative identity. In these instances, the individual must work to either incorporate a new (typically negative) experience into his or her existing self-story, or revise his or her narrative identity to accommodate the event. For example, researchers have examined associations between narrative identity and MH in the aftermath of divorce (King & Raspin, 2004), during a period of bereavement (Baddeley & Singer, 2010), and in the context of psychotherapy (Adler, Skalina, & McAdams, 2008). The findings from this literature have demonstrated that all narratives are not created equal and do not exert equal effects. Instead, different thematic and structural characteristics of narrative identity are differentially associated with MH (e.g., Adler, 2012; Adler, Chin, Kolisetty, & Oltmanns, 2012; Adler et al., in press; Adler & Poulin, 2009; Baddeley & Singer, 2010; Bauer & McAdams, 2010; Bauer, McAdams, & Sakaeda, 2005; Dunlop & Tracy, 2013; King & Raspin, 2004; King, Scollon, Ramsey, & Williams 2000; King & Smith, 2004; Lodi-Smith et al., 2009; Lysaker et al., 2012; Pals, 2006; Tavernier & Willoughby, 2012).

Although demonstrating the concurrent correlations between narrative characteristics and MH is a vital first step in supporting the assertions of narrative theory, it is also important to demonstrate that successful narrative identity supports MH over time, not just in the same moment. There are now a few longitudinal studies documenting this association over time. For example, Bauer and McAdams (2010) showed that different types of narratives about college students' life goals were differentially associated with MH outcomes 3 years after the narratives were written. In addition, Lodi-Smith and colleagues (2009) found that particular themes in college students' retrospective stories about their personality change during college were associated with increases in their emotional health over the period. Finally, Dunlop and Tracy (2013) found that adults enrolled in Alcoholics Anonymous whose story of their last drink featured the theme of redemption were more likely to report staying sober 4 months after writing the story, compared with participants whose narratives did not contain this theme. Studies like these serve to demonstrate the longitudinal associations between narratives collected at one time point and MH assessed at a separate time point.

Yet, like these three studies, virtually every study that has examined the longitudinal association between narrative identity and MH has relied on two-time-point designs. Although this lays an essential foundation, such designs are quite limited in their explanatory power for a number of reasons. Most importantly, two-time-point designs do not actually allow for a direct investigation of the key issue that most of these studies aspire to address: the nature of individual differences in intraindividual change. The central question in longitudinal research of this type is whether variation in narrative identity predicts change over time in MH. Demonstrating that such variation at Time 1 (T1) is associated with shifts between T1 and Time 2 (T2) does begin to address the matter, but, analytically, it often necessarily treats interindividual variability in change as error variance (see Collins & Sawyer, 2000, and Laurenceau, Hayes, & Feldman, 2007, for a further discussion of issues with two-time-point longitudinal designs). To evaluate the associations between variation in narrative identity and the shape and rate of change in MH over time, one needs more than two assessment points and different analytical tools.

There are two studies that have examined the relationship between narrative identity and MH over time using more than two assessment points, drawing on the same data set (Adler, 2012; Adler & Hershfield, 2012). In each study, a significant prospective association between particular themes in participants' narratives and their MH was observed. These two studies used psychotherapy as a vehicle for change in a wide range of life circumstances and, as such, were primarily focused on short-term change, with the average length of time between the first and final assessment point (usually the 12th assessment) being just under 15 weeks. There are a small number of longitudinal studies that have examined change over longer periods of time, but none permits an examination of the relationship between interindividual differences in narrative identity as they are associated with intraindividual trajectories of MH (or other outcomes). For example, McAdams and colleagues (2006) demonstrated the relative stability and variability in different narrative themes in a sample of emerging adults over 3 years, but they did not examine these differences in relation to MH or other outcomes, as that was not the aim of the study.

Thus, the two studies we report in this article provide the first evidence of the longitudinal association between variability in narrative identity and individual trajectories of MH over a substantial period of time. Each study includes a community sample of adults and draws on data collected at multiple time points over the course of several years. The unifying hypothesis underlying this pair of studies is that the associations that have been documented between different narrative themes and MH in cross-sectional and two-time-point longitudinal studies will also be affirmed when examining trajectories over several years. This association is central to the theory of narrative identity and remains untested using methodological designs and analytical tools best suited to addressing it.

## Developmental and Cultural Contexts

Both of the studies presented here include samples from late midlife. Study 1 includes participants in their mid- to late 50s, and Study 2 includes participants in their late 50s to early 60s. During this developmental period, one of the chief challenges is managing one's physical health (PH). Except in the case of unusually early mortality, PH issues are a universal and inescapable challenge—one that becomes increasingly common over the adult life course. Just as the biological changes of puberty help jumpstart identity development in adolescence (e.g., Erikson, 1959), biological decline throughout adulthood also prompts reflection on the past, present, and future (e.g., Kenyon, Bohlmeijer, & Randall, 2011). Of course, this developmental period is marked by a range of other common experiences that also have the potential to prompt a reexamination of one's sense of self, such as the transition to retirement and the transition to grandparenthood, among others. The present studies focus on the relationships between variation in narrative identity and trajectories of MH broadly, but highlight PH as one of the principal domains in which both narrative identity and MH are challenged during this developmental period.

There are many ways of approaching this reevaluation of one's identity in the face of such challenges. Theories of narrative identity suggest that people draw from a menu of scripts available in their culture for how lives ought to unfold (e.g., Berntsen & Rubin, 2004; Bohn, 2011; Habermas & Bluck, 2000; Hammack, 2008; McAdams, 2001; McAdams & Pals, 2006). Stories of people wrestling with PH challenges are abundant in contemporary American culture. For example, before his history of doping was uncovered, the story of cyclist Lance Armstrong's experience with cancer launched a high-profile public awareness campaign, selling over 47 million yellow "Live Strong" bracelets in its first year to support cancer research (Webster, 2005). The mass media also commonly takes the health struggles of everyday people and turns them into parables for popular consumption (e.g., Johnson, 1998). One of the reasons these stories may be so captivating is their ability to present a range of narrative options for individuals to use when it comes time to crafting one's own story of a PH challenge. These *master narratives* (Hammack, 2008) provide socially affirmed choices for how stories ought to unfold. As such, they are likely to be characterized by a set of themes that are widely available in contemporary American culture and that convey different options for how stories of PH challenges might sound. They also serve this purpose beyond the narrow specificity of physical decline, providing examples of general narrative strategies for making sense of hardship.

## Four Core Themes

The present pair of studies examines four narrative themes: agency, communion, redemption, and contamination. These themes were selected based on their ubiquity in American master narratives of navigating adversity, especially their relevance to narrating PH challenges and the strong traditions of empirical research demonstrating their associations with MH.

One of the central themes in contemporary American stories about PH is agency. "The fighter" is a common protagonist in illness narratives—a main character who goes to battle with his or her disease in a struggle for health and vitality. The essence of this character is his or her sense of personal agency—the ability to exert some influence over the course of his or her life, as opposed to being batted around by the whims of external forces. Agency is a central human concern (e.g., Bandura, 2006), taps the fundamental human need for autonomy (e.g., Deci & Ryan, 2000), and has been referred to as one of the overarching themes in life narratives (McAdams, Hoffman, Mansfield, & Day, 1996). There is a strong tradition of research linking the theme of agency in personal narratives to positive MH (e.g., Adler et al., 2008; Helgeson, 1994; Woike & Polo, 2001), including recent empirical data demonstrating that agency may increase prior to associated increases in MH (Adler, 2012). Indeed, agency has been examined in stories of a wide variety of challenging life experiences, including major life transitions (Bauer & McAdams, 2004), important and problematic moments in close relationships (Thorne & Michaelieu, 1996), and psychotherapy (Adler, 2012). In the context of PH challenges, there is mixed evidence that believing in one's ability to overcome one's illness is actually predictive of better PH outcomes (e.g., Coyne & Tennen, 2010), but there is convincing evidence that the personal story itself is associated with better MH and quality of life (e.g., Aspinwall & Teseschi, 2010; Chipperfield et al., 2012; Helgeson & Palladino, 2012; Petrie & Weinman, 2012).

The narrative theme of communion is often paired with agency as one of the superordinate themes in life narratives (e.g., McAdams et al., 1996). Moreover, communion often appears in stories about PH. Whereas some people respond to illness by adopting the guise of "the fighter," others focus on their social connections as the key to regaining vitality. Like agency, the drive for connection with others is a fundamental human need, described by Deci and Ryan (2000), for instance, as a need for relatedness. There is also a growing body of evidence linking the theme of communion, or social connectedness, to positive MH across a variety of contexts (e.g., Adler et al., 2012; Helgeson, 1994; Helgeson & Palladino, 2012; E. D. Mansfield & McAdams, 1996; Philippe, Koestner, Beaulieu-Pelletier, & Lecours, 2011; Woike & Polo, 2001). Illness not only operates within the individual but also sends ripples throughout his or her social network (e.g., Christakis, 2004; Kleinman & Seeman, 1999), and stories of connection are a common narrative option for responding to this challenge. Of course, highlighting one's relationships is also a beneficial way of narrating one's life experiences more broadly as well.

In addition to agency and communion, stories of adversity in contemporary American culture often display one of two trajectories: redemption or contamination. Redemption stories start out bad, but end good (e.g., McAdams, Reynolds, Lewis, Patten, & Bowman, 2001). McAdams (2013) suggested that redemption may be considered a signature theme in American history, heritage, and popular literature. Like canonical American stories of upward social mobility, rags to riches, and liberation from oppression, the narrative of recovery is an especially common redemptive story wherein the protagonist regains an early goodness, innocence, or health. Personal stories of recovery from illness may lead to personal growth, a reconceptualization of one's priorities, richer connections with others, or a deepening of one's spirituality. In each instance, the negative health experience is narrated as having led to a valued outcome. The positive end need not completely overwhelm the negativity of the health challenge to be understood as a redemption theme; the ability to glean some positivity amid the struggle can be considered redemptive enough to produce positive ends. Empirical research on redemption narratives has demonstrated their positive association with MH in a variety of domains (e.g., Adler & Poulin, 2009; Baddeley & Singer, 2008; Dunlop & Tracy, 2013; Lodi-Smith et al., 2009; McAdams et al., 2001).

In contrast to redemption sequences, contamination sequences follow the trajectory from good to bad. In these stories, scenes that start out positive are narrated as ending negatively. Whereas redemption sequences find a positive seed in the midst of negative experiences, the negativity in contamination sequences is described as overwhelming or polluting the preexisting positivity (e.g., McAdams et al., 2001). It is easy to see how stories of PH challenges might take a contaminative arc, with the onset of symptoms spoiling the health and vitality that preceded them. Contamination sequences have been shown to be associated with a range of poor MH outcomes (e.g., Adler, Kissel, & McAdams, 2006; Adler & Poulin, 2009; Baddeley & Singer, 2008; Lodi-Smith et al., 2009; McAdams et al., 2001).

These four themes—agency, communion, redemption, and contamination—represent four common master narratives of life's challenges, and each has a tradition of empirical research supporting its association with MH. Agency and communion are motivational themes, whereas redemption and contamination are affective themes, in narrative identity. Without a doubt, there are many other narrative themes—motivational, affective, and other types—that are associated with MH (e.g., Adler et al., in press), but these four warrant especially close attention in the current research because of their grounding in cultural master narratives and relevance to PH challenges.

Emerging research links all four themes to changes in MH over time. In a study by Adler (2012), the theme of agency in psychotherapy clients' stories increased over 15 weeks of treatment, and these increases temporally preceded associated increases in MH. In a study by Bauer and McAdams (2010), themes of communion in emerging adults' stories of their personal goals were associated with increases in MH 3 years later. In a study by Dunlop and Tracy (2013), Alcoholics Anonymous participants whose stories of their last drink included themes of redemption were more likely to report remaining sober 4 months later than those whose narratives did not contain this theme (Dunlop & Tracy, 2013). Finally, in a study by Lodi-Smith and colleagues (2009), low levels of themes of contamination (as part of a composite variable representing "affective processing") in college students' stories of their personality change during college were associated with increases in MH over the 4 years (Lodi-Smith et al., 2009). Thus, empirical evidence from longitudinal studies extends findings from cross-sectional research to demonstrate the association between the themes of agency, communion, redemption, and contamination and MH over time.

There is also an impressive array of findings from a distinct, but related, literature, focused on the expressive writing paradigm (e.g., Pennebaker, 1997). This body of research has demonstrated that writing about difficult experiences has significant associations with both PH and MH (e.g., Pennebaker & Chung, 2011). Although longitudinal research has been conducted beyond the immediate aftermath of the intervention, data suggest that the beneficial results of this approach for both MH and PH are maintained up to 6 months later (Pennebaker & Chung, 2011). Despite some conceptual overlap, research on the expressive writing paradigm bears important differences from research on narrative identity that prevents its conclusions from directly translating to the present pair of studies. Most centrally, research on expressive writing has primarily employed linguistic, as opposed to narrative, approaches to examining text. Linguistic approaches emphasize the role of specific categories of word use, whereas narrative approaches focus on the emergent meaning in personal stories, which is composed of, but irreducible to, the specific words used (e.g., Weston, 2014). The scant empirical research that has directly examined the differences between linguistic and narrative components of text has demonstrated little overlap in their prediction of health outcomes (e.g., Graybeal, Sexton, & Pennebaker, 2002).

## The Present Pair of Studies

The present pair of studies employs different longitudinal designs that include multiple assessment points to examine a core assertion of the theory of narrative identity that remains untested: that different ways of constructing a personal narrative ought to predict different trajectories of MH over time. Although existing research has documented the association between the themes of agency, communion, redemption, and contamination in a variety of cross-sectional contexts and a few two-time-point longitudinal designs, no study has allowed for a direct investigation of the central longitudinal question. One of the key assertions of the theory of narrative identity is that personal narratives do not just serve a psychological function in the moment that they are constructed or told, but that the way a personal narrative thematically and structurally instantiates the narrator's sense of purpose and unity should serve as a foundation for supporting his or her MH over time (e.g., Dunlop & Tracy, 2013; Giddens, 1991; McAdams, 2001). The existing evidence supports the conclusion that interindividual variability in narrative identity is associated with MH, but there are no data that can speak to whether the association between these interindividual differences is associated with intraindividual trajectories of MH over substantial periods of time. The present studies provide two approaches to this question, relying on methodological designs and analytical strategies employed to directly investigate this question. We have selected samples with common developmental concerns, one of which is challenges to PH, in examining the matter. Both studies share the core hypotheses that higher levels of the themes of agency, communion, and redemption, and lower levels of contamination will be associated with positive trajectories of MH, over time.

## Study 1

Study 1 investigated the relationship between individual differences in narrative identity and intraindividual differences in trajectories of MH over the course of 4 years in a community sample of late-mid-life adults. As such, this study can be understood as an initial test of the key questions outlined earlier under naturalistic circumstances, wherein the sample participants were followed over time and their MH was periodically reassessed.

The participants in Study 1 were drawn from a larger investigation of narrative identity in late-mid-life. Of relevance to the present study, they participated in an extensive life story interview designed to collect not only core components of the typical life story interview but also several specific topics relevant to participants in this developmental period (McAdams, 2008). The interview included an unusual section, wherein participants were asked to tell the story of a significant health challenge. Thus, the data from this study allow not only for an investigation of overall trends in the association between narrative identity and trajectories of MH but also a focused examination of the thematic content of health challenge stories and MH in a sample for whom this should be a salient present concern.

Based on the conclusions from the accumulated body of research on narrative identity and MH that has employed cross-sectional and two-time-point longitudinal designs, the hypotheses of Study 1 are that higher levels of agency, communion, and redemption, and lower levels of contamination, in the narratives of late-mid-life adults will be associated with positive trajectories of MH over the subsequent 4 years. These themes ought to be associated with increases—as opposed to declines or stability—in MH over time. The sample also permits a focused investigation of these hypotheses using data from a chapter in the life story specifically describing the participant's greatest health challenge—a concern that should be relevant in the current lives of participants in the sample. There is no theoretical or empirical basis on which to develop a separate set of hypotheses concerning health narratives, but additional analyses will examine whether the general hypotheses are supported when these specific stories are used. The data therefore permit an investigation of whether the association between narrative themes and trajectories of MH are specific to certain types of stories or are identified when examining personal narratives more generally.

## Method

**Participants.** Participants were drawn from a larger, ongoing, 10-year longitudinal study of aging in the baby boomer birth cohort (see Jones & McAdams, 2013, for a more thorough description of the broader study methods). All measures in the present study were drawn from the first five assessment points in this study: at the time of enrollment (Baseline) and once a year for the next 4 years. Participants were recruited by an independent private research firm through print advertisements and mailings. Interested individuals contacted the research firm by telephone and were asked a series of screening questions. To qualify for the broader study, individuals must have been born during the years of 1951 to 1954, identify as either African-American/Black or Euro-American/White, and have earned at least a high school diploma. The broader study was designed to investigate questions pertaining to race and therefore aimed to include roughly equal numbers of African-American/Black and Euro-American/White participants. As a result, some targeted recruitment of African-American/Black participants through advertisements in media focused on African-Americans was conducted by the research firm.

The sample for the present study was 89 adults between the ages of 55 and 59 at Baseline. In order to permit investigation of the unique association between narrating one's greatest health challenge and one's trajectory of MH, participants were selected for inclusion in the present study if they discussed a personal health challenge, as opposed to a health challenge faced by a family member or close other during the life story interview (representing 54% of the broader data set). Analyses comparing the sample with the larger data set from which it was drawn indicated that the sample did not significantly differ from the unselected participants in terms of age, $t(161) = -.01$, $p = .99$, race, $t(161) = .98$, $p = .33$, education, $t(161) = .72$, $p = .47$, income, $t(161) = .03$, $p = .98$, or scores on the MH and PH composites described in the following sections (MH: $t(161) = 1.46$, $p = .15$; PH: $t(161) = -.088$, $p = .37$). The sample did include proportionally more men than the broader data set, $t(161) = 2.82$, $p < .001$. Forty-nine (55%) of the participants were male and 43 (42%) were African-American/Black. The average level of education in the sample was a college degree. Demographic characteristics of the sample are presented in Table 1. As a result of efforts by the broader study team to retain participants, attrition was limited. In the present sample, of the 89 participants at Baseline, 88 provided data at T2 (1 year after Baseline), 87 provided data at Time 3 (2 years after Baseline), 85 provided data at Time 4 (3 years after Baseline), and 83 provided data at Time 5 (4 years after Baseline). Because of this small degree of attrition, no analyses were undertaken to determine whether the dropouts meaningfully differed from the rest of the sample.

**Measures.**

***MH.*** MH was assessed via four widely used questionnaires: the Psychological Well-Being scales (Ryff & Keyes, 1995; Cronbach's alpha = .87), the Social Well-Being scales (Keyes, 1998; Cronbach's alpha = .72), the Satisfaction with Life Scale (Diener, Emmons, Larsen, & Griffin, 1985; Cronbach's alpha = .83), and the Beck Depression Inventory-II (BDI-II; Beck, Steer, & Brown, 1996; Cronbach's alpha = .82). These four measures were selected to tap a broad-based assessment of MH. Scores on the Baseline assessment of the four measures were highly intercorrelated (absolute value of Pearson $rs = 0.37$ to 0.63), so a composite variable representing MH was created, with higher scores indicating better MH.

***PH.*** PH was assessed using the SF-12 (Ware, Kosinski, & Keller, 1994; Cronbach's alpha = .88), one of the most widely used measures of self-reported PH, and a self-report checklist indicating whether the individual had ever been diagnosed with

Table 1
*Demographic Characteristics of Samples*

| | Study 1 ($n$ = 89) | | Study 2 | | | |
|---|---|---|---|---|---|---|
| | | | Illness group ($n$ = 27) | | Control group ($n$ = 27) | |
| | Number (% of sample) | *M* (*SD*) | Number (% of sample) | *M* (*SD*) | Number (% of sample) | *M* (*SD*) |
| Sex | | | | | | |
| Female | 49 (55) | — | 13 (48) | — | 13 (48) | — |
| Male | 40 (45) | — | 14 (52) | — | 14 (52) | — |
| Race | | | | | | |
| African-American/Black | 43 (48) | — | 6 (22) | — | 6 (22) | — |
| Euro-American/White | 46 (52) | — | 21 (78) | — | 21 (78) | — |
| Age | — | 56.38 (1.48) | — | 60.22 (2.61) | — | 60.26 (2.65) |
| Income | — | 5.56 (4.26) | — | 2.74 (1.68) | — | 2.76 (1.98) |
| Education | — | 3.01 (1.02) | — | 5.96 (2.12) | — | 5.96 (2.12) |

*Note.* Study 1 – Income: 1 = <\$25,000/year, 2 = \$25,000–\$49,999/year, 3 = \$50,000–\$74,999/year, 4 = \$75,000–\$99,999, 5 = \$100,000–\$124,999, 6 = \$125,000–\$149,999/year, 7 = \$150,000–\$174,999/year, 8 = \$175,000–\$199,999/year, 9 = \$200,000–\$224,999/year, 10 = \$225,000–\$249,999/year, 11 = \$250,000–\$274,999/year, 12 = \$275,000–\$300,000/year, 13 = >\$300,000/year; Study 1 – Education: 1 = high school, 2 = some college, 3 = college degree (BA or BS), 4 = graduate work (postcollege), 5 = advanced degree; Study 2 – Income: 1 = <\$20,000/year, 2 = \$20,000–\$39,999/year, 3 = \$40,000–\$59,999/year, 4 = \$60,000–\$79,999/year, 5 = \$80,000–\$99,999/year, 6 = \$100,000–\$119,999/year, 7 = \$120,000–\$139,999/year, 8 = >\$140,000/year; Study 2 – Education: 0 = none, 1 = elementary or junior high, 2 = GED, 3 = high school diploma, 4 = vocational/technical degree, 5 = associate's degree, 6 = RN diploma, 7 = bachelor's degree, 8 = master's degree, 9 = doctoral degree.

each of 34 medical conditions, drawn from the Midlife in the United States study (Ryff & Almeida, 2009). The score on the Baseline SF-12 was highly correlated with the Baseline sum of the medical diagnoses ($r = .48, p < .001$), so a composite of these measures representing PH was created, with higher scores indicating better PH.

***Narratives.*** Participants engaged in a Life Story Interview (McAdams, 2008), a 2- to 3-hr semistructured interview developed and widely used for collecting full accounts of individuals' lives. Interviews were audio recorded and professionally transcribed, verbatim. The interviews produced a very large amount of text ($M$ = 17,216 words per participant, $SD$ = 5,521). For the purpose of the current study, four core scenes were selected for investigation: scenes describing participants' high points, low points, turning points, and greatest health challenge. For each scene, participants were asked to describe the event in detail, including when and where it took place, who was there, and what they were thinking and feeling, and to discuss what the scene may say about who they are as a person. Each scene was quite substantial in length (high points: $M$ = 879 words, $SD$ = 494 words; low points: $M$ = 1,010 words, $SD$ = 576 words; turning points: $M$ = 958 words, $SD$ = 476 words; health challenge: $M$ = 760 words, $SD$ = 551 words). The health challenge chapters were extremely varied in content, covering topics including near-fatal brushes with cancer, recovery from major car accidents, acute infections, and struggles with chronic problems such as back pain, obesity, diabetes, and asthma.

Once each year for 4 years, following the life story interview and initial battery of self-report questionnaires, participants again completed the measures included in the MH and PH composite variables, allowing for assessment of their trajectories over time.

***Coding of the narratives.*** Participants' narratives were rated by reliable coders using widely used, reliable, and valid coding systems for the themes of agency, communion, redemption, and contamination. Following typical practice in narrative research, the complete set of narratives was coded for a given theme before moving on to the next theme. Coders were trained to stringent standards of interrater reliability with each coding system before undertaking the coding process in earnest, working iteratively with subsets of the data in collaboration with the first author.

The theme of agency was rated using a 4-point scale drawn from previous studies that examined narrative identity and MH (e.g., Adler, 2012; Adler et al., 2012). In this system, a "0" indicates that the individual portrayed himself or herself as completely powerless, at the mercy of the whims of fate or other external forces. For example, in discussing her ongoing struggles with obesity, one participant said, "I feel claustrophobic, like I'm *in* somebody else, like the fat is smothering me. It's awful." She portrayed her excess body fat as powerfully acting upon her, nearly robbing her of her own identity. In contrast, a "3" was given when the individual portrayed himself or herself as having been the major driver of the events described. For example, one participant described an insight she had while taking a course in college that introduced her to feminist theory:

> A lot of people that were in that course were women that were coming back into the college world. I was just starting out. And it was kind of a neat class because I got to see from their experience how they were dependent on the, the men in their life . . . And it was like, "Well, yeah, who are you?" And, you know, I'm thinking, "Do you have an identity?" And that was the first thing for me: I am going to keep my identity. I'm going to learn who I am, and I am going to be a strong woman. And I think that was a big point in my life where I saw how important it was to be my own person and create my own identity even to the point when I got married, I didn't want to give up my maiden name because I felt like, no, I don't want to lose this. This is who I am, and I will add on my married experience to it. And thankfully enough, it was a short enough name that it didn't—wasn't too much of a hardship on, on anything after that. But, that meant a lot to me.

This participant described how seeing the examples of her classmates led to her own determination "to be a strong woman" and how that manifested in her life over time. She recalls this experience as a key turning point in her life story, one that establishes her as a highly agentic person. In the present study, two coders who were blind to the hypotheses and participant characteristics rated the narratives with excellent interrater reliability (Intra-class correlation (ICC) Case 2, Type I = 0.82; Shrout & Fleiss, 1979). For analyses, scores across the scenes were averaged or, in the case of targeted analyses by scene, used independently.

The theme of communion was rated using a 4-point scale drawn from previous studies that assessed life stories (e.g., Adler et al., 2012). In this system, a "0" indicates that the narrative is lacking in any talk of motivation for connection or explicitly discusses motivation for disconnection. For example, in describing her decision to have a fibroid tumor removed, one participant said,

> It was a big decision. It was only my decision. That is one of those things that—it is not like you could share that decision with anybody else, and I did not share it. I knew my husband's view on doctors which is, it would have been, never do it, never do it. It was just my decision and I made it.

In this excerpt, the participant explains how she was motivated to make the decision without any other input, even from her husband, thus demonstrating low levels of communion. In contrast, a "3" was given when there is a high degree of discussion of interpersonal relationships and the individual described his or her motivation for close connection with others. For example, here is an excerpt from one participant's discussion of her experiences in the wake of a stroke:

> I had the stroke and . . . it gives you so much fear and I was terrified at that. But even with the fear it was like spirits [visited me]. My great-grandson came over and cooked me breakfast. I had some rubber eggs and some burnt bacon, and some grits that I don't know what he did to them [laughs]. He cooked me breakfast and it took him three hours to do it 'cause I think he maybe messed up the first few times, you know. He was instructed he's not to let me out of bed so that was a good experience during that time . . . But having gone through all of that, even with the rubber eggs . . . that's one of them times you, you learn that I don't care how little family you got, friends, coworkers, or whatever, when you get in a position like that they can help you.

In this example, the participant makes clear how her family, like "spirits," swooped in to take care of her while she recovered and how meaningful that was to her. Two coders, blind to the hypotheses, coded the narratives with excellent interrater reliability (ICC Case 2, Type I = 0.82). For analyses, scores across the scenes were averaged or, in the case of targeted analyses by scene, used independently.

Redemption and contamination were rated using a system developed by McAdams and colleagues (2001) that has been used in a wide range of previous studies (e.g., Adler & Poulin, 2009; Lodi-Smith et al., 2009; McAdams et al., 2001). In this coding system, each narrative scene is assigned a score of 0 (*absent*) or 1 (*present*) for each theme. Redemption is rated as present when a narrative shows a clear shift from an undesired or negative beginning to a subsequent positive ending. The beginning and ending must be narrated as connected to each other and the emergent positivity must somehow undo the core (if not the entirety) of the initial negativity. For example, one participant told the story of becoming pregnant as a teenager, the low point of her life. She starts the story on the day she came home from high school, knowing that her school counselor was going to call and inform her mother:

> I *couldn't* tell my mother, and I knew the phone was going to ring. I went and got in the bathtub. I *could not* tell my mother. And I heard my mother say "What?!" . . . I was scared. I didn't know what to do. I didn't know, I didn't know nothing about birth control or nothing. I didn't know anything . . . and I always say, although it was the lowest point, it was a blessing because I feel that God knew before I knew. As I always said, if I wouldn't have had that child then I would not have a child today 'cause I haven't met anybody that's worth anything. So God blessed me. That was a blessing for me, and I know my son loves me . . . that was the lowest point, and it came out to be a blessing.

She describes how this nadir experience resulted in a relationship she is deeply grateful for, thus redeeming the terror and shame of the day her mother found out.

Contamination is rated as present when a narrative shows a clear shift from a pleasant or positive beginning to a subsequent negative ending. The beginning and ending must be narrated as connected to each other and the negative state at the end must spoil or overcome the previous positive state. For example, one participant described the onset of his illness in his early 50s:

> I usually feel very unified. I think I've always sort of felt unified and for the first time I didn't feel unified. It's scary, you know, and I thought, it truly, it was the kind of thing where for the first time I couldn't say, "For 50 years I've been fine. I'm going to be fine again." It was, it was sort of like, "Oh my gosh, this has happened and I can't see how this will ever end and I can't get out of it." I mean it, it's you know, like if your knee always hurts, you can at least go about your business. But this was debilitating.

The experience made this participant feel vulnerable and disconnected for the first time in his life, destroying his historically solid sense of himself.

Narratives may receive scores noting the presence of both redemption and contamination. Coefficient kappa was used as the index of interrater reliability for redemption and contamination coding. Two coders, blind to the hypotheses, achieved excellent interrater reliability ($\text{kappa}_{\text{redemption}} = 0.80$; $\text{kappa}_{\text{contamination}} = 0.83$). For analyses, scores across the scenes were summed or, in the case of targeted analyses by scene, used independently. Thus, the metric for all four themes was 0 to 4 when all four scenes were examined as a set.

## Results

**Baseline cross-sectional analyses.** Zero-order correlations presenting the cross-sectional relationships between Baseline demographic, narrative, and MH and PH variables are presented in Table 2. There were few significant relationships between demographic variables and the narrative variables and MH or PH (see Table 2). There were almost no significant associations among the narrative themes, with the exception of agency and redemption, which were significantly positively correlated, $r = .26$, $p < .05$. It is important to note that the length of the narratives was not associated with any of the narrative variables or demographic variables, or with MH or PH.

These correlational analyses provided support for most of the first set of hypotheses in the present study, as they indicate that at the concurrent time point, MH, but not PH, was significantly correlated with the narrative themes (assessed across all four scenes) of agency (MH: $r = .34$, $p < .01$; PH: $r = -0.05$, $p = .74$), redemption (MH: $r = .33$, $p < .01$; PH: $r = .05$, $p = .65$), and contamination (MH: $r = -0.36$, $p < .01$; PH: $r = -0.18$, $p = .08$). However the theme of communion was not significantly associated with MH or PH (MH: $r = .14$, $p = .20$; PH: $r = .14$, $p = .19$). Simultaneous regression equations including all four narrative themes produced analogous results (MH: $R^2 = 0.26$, $\beta_{\text{agency}} = 0.22$, $p < .05$; $\beta_{\text{communion}} = 0.08$, $p = .46$; $\beta_{\text{redemption}} =$

Table 2
*Study 1: Intercorrelations Between Time 1 Variables*

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1. Sex | | | | | | | | | | | |
| 2. Race | −.06 | | | | | | | | | | |
| 3. Age | .03 | −.16* | | | | | | | | | |
| 4. Income | −.02 | .12 | −.16* | | | | | | | | |
| 5. Education | −.09 | −.11 | −.10 | .21** | | | | | | | |
| 6. Agency | .19 | .08 | .12 | −.13 | .18 | | | | | | |
| 7. Communion | .03 | −.24* | .08 | .06 | −.11 | −.01 | | | | | |
| 8. Redemption | .18 | .14 | −.17 | −.04 | −.01 | .26* | −.09 | | | | |
| 9. Contamination | −.06 | .11 | −.16 | −.18 | −.01 | −.16 | −.20 | −.04 | | | |
| 10. Word count | .11 | .01 | −.15 | .01 | .19 | .08 | .11 | .05 | .15 | | |
| 11. Physical health | −.06 | −.02 | −.12 | −.16* | −.13 | .04 | .14 | −.05 | −.18 | −.11 | |
| 12. Mental health | .19* | −.12 | .18* | .12 | .03 | .34** | .14 | .33** | −.36** | .03 | −.01 |

* $p < .05$. ** $p < .01$.

0.24 $p < .05$; $\beta_{contamination} = -0.29$, $p < .01$; PH: $R^2 = 0.06$, $\beta_{agency} = 0.05$, $p = .69$; $\beta_{communion} = 0.11$, $p = .32$; $\beta_{redemption} = -0.05$, $p = .68$; $\beta_{contamination} = -0.14$, $p = .23$). These findings largely replicate those from previous research that has documented the associations between three of these narrative themes and MH.

**Longitudinal analyses.** The primary analytical approach used for examining longitudinal trends was growth curve modeling, also called hierarchical linear modeling (HLM; e.g., Bryk & Raudenbush, 1987; Garson, 2012; Nezlek, 2011). HLM is well suited to the analysis of real-world longitudinal data because it can accommodate missing or unevenly spaced data points in modeling the growth curves of variables (e.g., Singer & Willett, 2003). The first set of longitudinal models from this study is presented in Table 3. In the present study, there were no general trends over time in participants' PH or MH when examined across the entire sample (see Models 1 and 2). Given the diversity of the sample's life experiences, this lack of clear overarching patterns was not surprising.

Participants' narrative themes across all four scenes were unrelated to their PH trajectory over the course of 4 years (Model 3). In contrast, participants' trajectories of MH showed significant variation based on individual differences in their narrative themes across all four scenes, even after statistically accounting for the passage of time (Model 4). Specifically, participants whose narratives featured higher levels of the themes of agency and redemption had more positive trajectories of MH over the subsequent 4 years compared with those with lower levels of these themes, whereas participants whose narratives featured higher levels of contamination had more negative trajectories of MH over 4 years than those with lower levels. There were no reliable differences in MH trajectory based on participants' theme of communion. The addition of demographic variables into the model did not significantly impact the findings. Furthermore, a model that included participants' trajectory of PH over time did not significantly impact the findings (Model 5), indicating that the relationship between intraindividual variation in narrative identity and trajectories of MH over time was not impacted by participants' PH over time.

**Exploratory analyses, by scene.** Given the unusual inclusion of a prompt in the life story interview conducted with this sample that asked participants to describe their greatest health challenge, as well as the developmental relevance of this experience for the sample, a set of exploratory analyses was conducted to determine whether individual differences in the narration of this scene was associated with different trajectories of MH or PH. In narrative research, it is somewhat unusual to conduct analyses using a single scene when more are available, barring theoretical predictions about the uniqueness of specific stories. For example, C. Mansfield, McLean, and Lilgendahl (2010) developed specific hypotheses about differences between narratives of traumas and narratives of transgressions. In their analyses, they examined these single-scene narratives separately and found distinct associations between their thematic content and well-being. In the present study, PH challenges are developmentally relevant, given the

Table 3
*Study 1 Growth Curve Models (Final Estimation of Fixed Effects With Robust Standard Errors)*

| Parameter | Coefficient | *SE* | *t* |
|---|---|---|---|
| Model 1: Physical health over time | | | |
| Intercept | 0.68 | 0.02 | 0.95 |
| Slope | −0.02 | 0.04 | −0.54 |
| Model 2: Mental health over time | | | |
| Intercept | −0.10 | 0.05 | −0.15 |
| Slope | −0.01 | 0.01 | −0.37 |
| Model 3: Physical health over time with narrative themes at Level 2 | | | |
| Intercept | 0.65 | 0.07 | 0.94 |
| Slope | −0.02 | 0.04 | 0.60 |
| Agency | 0.02 | 0.33 | 0.06 |
| Communion | 0.27 | 0.31 | 0.88 |
| Redemption | 0.49 | 0.71 | 0.69 |
| Contamination | −1.15 | 0.88 | −1.31 |
| Model 4: Mental health over time with narrative themes at Level 2 | | | |
| Intercept | 0.01 | 0.01 | −0.39 |
| Slope | 0.44 | 0.19 | 2.37* |
| Agency | 0.17 | 0.09 | 1.92* |
| Communion | 0.11 | 0.09 | 1.29 |
| Redemption | 0.25 | 0.17 | 2.29* |
| Contamination | −0.51 | 0.27 | −1.97* |
| Model 5: Relationship between mental health and physical health over time with narrative themes at Level 2 | | | |
| Intercept | −0.10 | 0.14 | −0.67 |
| Mental health slope | 0.37 | 0.18 | 1.91* |
| Physical health slope | −0.01 | 0.10 | −0.66 |
| Agency | 0.08 | 0.04 | 2.03* |
| Communion | 0.01 | 0.05 | 0.83 |
| Redemption | 0.17 | 0.08 | 1.87* |
| Contamination | −0.17 | 0.09 | −1.86* |

* $p < .05$.

late-mid-life sample. Exploratory analyses compared individual differences in the narration of health challenges with narratives of high points, low points, and turning points in their association with trajectories of MH and PH over 4 years.

Baseline cross-sectional analyses indicated that the general associations between the narrative themes and MH and PH were observed at the level of specific scenes only for narratives of health challenges and of low points (see Table 4). In other words, individual differences in the themes of agency, redemption, and contamination were significantly associated with MH in health challenge and low-point scenes, but not in high-point and turning-point scenes. As in the analyses of general trends, no significant associations were observed for any narrative theme in any scene and PH (though there was a marginally significant negative association between contamination in low-point scenes and PH), or between the theme of communion in any scene and MH.

Longitudinal analyses also indicated that the patterns observed when examining the associations between individual differences in narrative themes across all scenes and trajectories of MH were generally replicated when themes were examined within health challenge and low-point scenes, but not within high-point or turning-point scenes (see Models 1 through 4 in Table 5). In other words, individual differences in the themes of agency, redemption, and contamination in health challenge and low-point scenes—but not in high-point and turning-point scenes—were significantly associated with trajectories of MH (but not PH). (The one exception was an additional significant result for contamination in turning-point scenes and trajectories of MH.) No significant associations were observed between individual differences in communion and trajectories of MH (though a marginally significant association was observed between individual differences in communion in turning-point scenes and trajectories of MH). Taken together, the results of the baseline cross-sectional analyses and the longitudinal analyses suggest that there may be differential associations between narrative themes and MH, both concurrently and prospectively, depending on the scene that is narrated.

Table 4
*Study 1: Correlations Between Time 1 Themes and Outcomes by Scene*

| | Physical health $r$ | Mental health $r$ |
|---|---|---|
| Agency | | |
| Health challenge | .12 | $.27^{**}$ |
| Low point | −.07 | $.28^{***}$ |
| High point | .05 | .17 |
| Turning point | −.19 | .02 |
| Communion | | |
| Health challenge | .08 | −.10 |
| Low point | .07 | .18 |
| High point | −.01 | .02 |
| Turning point | .18 | .17 |
| Redemption | | |
| Health challenge | .16 | $.22^{**}$ |
| Low point | −.05 | $.37^{***}$ |
| High point | .05 | −.10 |
| Turning point | −.10 | −.15 |
| Contamination | | |
| Health challenge | −.15 | $-.25^{**}$ |
| Low point | $-.21^{*}$ | $-.28^{***}$ |
| High point | .17 | .02 |
| Turning point | −.06 | −.17 |

$^{*}p = .07.$ $^{**}p < .05.$ $^{***}p < .01.$

## Discussion

The results of Study 1 provide two general conclusions about the relationship between narrative identity and MH. First, when assessed concurrently, the findings from this study add to the growing body of research that has identified significant associations between the themes of agency, redemption, and contamination and MH. Second, the findings from this study provide the first data to suggest that individual differences in narrative identity are associated with different trajectories of MH over a substantial period of time. In this sample of late-mid-life adults, participants whose narratives included higher levels of the themes of agency and redemption and lower levels of the theme of contamination experienced more positive trajectories of MH over 4 years compared with participants whose narratives had different thematic content. These findings were largely consistent across participants' age, race, education level, and income level. There were no significant relationships between individual differences in participants' narratives and their trajectories of PH over 4 years. This finding aligns with theoretical work on narrative identity that distinguishes between the objective reality of one's lived experiences and the storied meaning that is constructed from those experiences (e.g., McAdams, 2001).

The pattern of results from Study 1 suggests that narrating one's experiences with a greater sense of agency and with the theme of redemption—and without the theme of contamination—may support MH over time. One excerpt from a participant's narrative makes this especially clear. In concluding the story of her significant health challenge, this participant said,

> I value that experience in that it (along with the experiences of my aunt) has given me a positive and healthy outlook . . . I've often said to myself through my life that if I'm ever told that I have cancer that I'm going to say to the doctor, "Okay, so what's next? What do we do now? Because I'm not trying to die. What are we going to do to eradicate this cancer? What are my options?" That's the position I'm going to take. It doesn't matter what type of disease, cancer, whatever . . . I have personal experiences that I can look at that say it's possible that I can get through that if that should ever happen in my life.

In this passage, the participant demonstrates how her highly agentic and redemptive narrative of her past health challenge has given her a foundation for approaching hypothetical future challenges that may arise. She says that she will feel empowered the next time she gets sick and values the health challenge for its ability to demonstrate how positive outcomes may result from negative experiences. Even among participants whose narratives do not draw such explicit connections between their stories of the past and their anticipated future selves, the results from this study suggest that thematic elements of narrative identity differentially relate to prospective trajectories of MH.

The lack of significant findings in Study 1 regarding the theme of communion was unexpected. Previous theoretical and empirical work has pointed to the positive association between a drive for connection with others and positive MH. However, as Helgeson and Palladino (2012) point out, questionnaire-based measures of communion have shown inconsistent associations with MH and PH. Communion scores were somewhat low in the present sample,

Table 5
*Study 1 Growth Curve Models (Final Estimation of Fixed Effects With Robust Standard Errors) by Scene*

| Parameter | Coefficient | *SE* | *t* |
|---|---|---|---|
| Model 1: Mental health over time with agency by scene at Level 2 | | | |
| Intercept | −0.01 | 0.01 | −0.40 |
| Slope | 0.33 | 0.15 | 2.24** |
| Agency health challenge | 0.08 | 0.05 | 1.75** |
| Agency low point | 0.10 | 0.05 | 2.09** |
| Agency high point | 0.07 | 0.06 | 1.17 |
| Agency turning point | −0.02 | 0.08 | −0.24 |
| Model 2: Mental health over time with communion by scene at Level 2 | | | |
| Intercept | −0.14 | 0.15 | −0.91 |
| Slope | 0.34 | 0.11 | 2.06** |
| Communion health challenge | −0.01 | 0.05 | −0.22 |
| Communion low point | 0.07 | 0.04 | 1.64 |
| Communion high point | −0.03 | 0.05 | −0.67 |
| Communion turning point | 0.10 | 0.04 | 1.69* |
| Model 3: Mental health over time with redemption by scene at Level 2 | | | |
| Intercept | −0.13 | 0.08 | −1.60 |
| Slope | 0.31 | 0.11 | 1.41** |
| Redemption health challenge | 0.17 | 0.08 | 2.00** |
| Redemption low point | 0.31 | 0.03 | 3.42*** |
| Redemption high point | −0.02 | 0.08 | −0.26 |
| Redemption turning point | −0.08 | 0.09 | −1.01 |
| Model 4: Mental health over time with contamination by scene at Level 2 | | | |
| Intercept | −0.12 | 0.07 | −1.63 |
| Slope | 0.28 | 0.09 | 1.93** |
| Contamination health challenge | −0.26 | 0.09 | −1.98** |
| Contamination low point | −0.18 | 0.10 | −1.77** |
| Contamination high point | 0.01 | 0.40 | 0.02 |
| Contamination turning point | −0.29 | 0.15 | −2.00** |

* $p = .09$. ** $p < .05$. *** $p < .01$.

though population means are unknown. On a scale of 0 to 3, the means and standard deviations for each scene were as follows: health challenge, $M = 1.30$, $SD = 1.02$; high point, $M = 2.21$, $SD = 0.96$; low point, $M = 1.01$, $SD = 0.98$; turning point, $M = 1.44$, $SD = 0.97$. It is possible that there was less variation in communion in the present sample than in other samples, and that this drove the nonsignificant results. It is also possible that more specific coding of communion, looking at contextualized communion with particular people or particular types of social connections, might have yielded significant results.

The results of exploratory analyses indicate that both the cross-sectional and prospective associations between narrative themes and MH may vary depending on the type of scene narrated. Specifically, the overall trends were largely replicated when examining scenes in which participants narrated a significant health challenge or their life's low point, but not in scenes describing their life's high point or turning point. Given that narratives of a significant health challenge demonstrate a similar pattern of associations with MH as low points, this suggests that the shared focus on negative events common to narratives of both health challenges and low points may be the key ingredient in their association with MH. Although the valence of the events described was not systematically coded, anecdotal reports of the coders indicated that the vast majority of health-challenge and low-point scenes contained negative events, whereas extremely few high-point scenes did. Turning-point scenes were quite varied, with some including negative events, but many simply describing a shift from one positive state to another (such as the transition to parenthood or the shift from student to professional). This variability may explain the lack of results for turning-point scenes (also see Tavernier & Willoughby, 2012). As noted earlier, there is little research that has directly examined the matter of scene content in the association with MH over time. Without a doubt, the results from scene-level analyses ought to be interpreted as preliminary. Single codes (especially for themes of redemption and contamination, which were coded dichotomously) are less robust predictors of trajectories of change than codes aggregated across several scenes. Nevertheless, these results open a promising avenue for future investigation.

## Study 2

The central question in Study 2 was whether the associations between individual differences in narrative identity and intraindividual trajectories of MH would also be identified in the wake of a specific negative life experience: the onset of a major physical illness. Whereas Study 1 sought to identify patterns in a naturalistic longitudinal sample, Study 2 investigated these same relationships in the context of a narrowly defined natural experiment. The sample for Study 2 was drawn from a much larger data set, that allowed for the creation of two groups, matched on demographic characteristics—one that contained participants who experienced the onset of a major physical illness at a precise time point in the series of longitudinal assessments, and the other that contained participants who remained healthy. This approach, although sacrificing some ecological validity, provides a more stringent test of the general hypotheses examined in Study 1. It is also unique in the narrative literature, though a study by Tavernier and Willoughby (2012) used a matched sample design in a two-time-point study of adolescents' turning-point stories.

The hypotheses of Study 2 are the same as in Study 1. First, we hypothesize that the concurrent relationships between the narrative themes of agency, communion, redemption, and contamination and MH will be replicated in this new and unusual sample. Second, the longitudinal associations between variability in the narrative themes and trajectories of MH will be tested to determine whether they replicate across the entire new sample, or only among participants who experience the onset of a physical illness. The design of Study 2 allows for the testing of these hypotheses in two separate groups of participants—one with a shared negative life experience and a control group of sorts, without a specifically shared life challenge. Thus, this study is centrally concerned with whether variation in narrative identity is associated with different trajectories of MH in the wake of a specific negative experience.

## Method

**Participants.** Participants in the present study were drawn from a broader study of personality and aging (see Oltmanns, Rodrigues, Weinstein, & Gleason, 2014, for a detailed description of study methods, including participant recruitment and retention). From the large (over 1,500 participants), epidemiologically representative, community-based sample, a select group of participants were identified whose PH time course allowed for a close examination of the association between narrative identity and trajectories of MH in the wake of a major illness diagnosis. Every participant from the broader sample who satisfied three criteria was selected: (a) he or she was free of significant PH diagnoses through the Baseline assessment point; (b) he or she reported being diagnosed with a serious illness between Baseline and T2, 6 months later; and (c) he or she did not report receiving any additional significant medical diagnoses between T2 and T5, 2 years following Baseline. In other words, participants were selected who had been mostly healthy up until receiving a serious illness diagnosis between Baseline and T2, and then did not receive any additional illness diagnoses for the duration of the study. These criteria were adopted in order to allow the results to most directly speak to the role of narrative identity in adapting to a specific subsequent illness diagnosis.

Twenty-seven participants from the broader sample met these three criteria and were selected for inclusion in the present study (illnesses diagnosed between Baseline and T2, 6 months later, were as follows: arthritis, $n = 4$; cancer, $n = 6$; cardiac illnesses, $n = 5$; diabetes, $n = 6$; gastrointestinal illnesses, $n = 4$; and pulmonary diseases, $n = 2$). It is important to note that this particular sample was not selected to be representative of the population as a whole, or the population of people who receive serious illness diagnoses; they were selected in order to address the specific longitudinal hypotheses in Study 2, and so reducing the number of confounding variables was prioritized over external validity.

Using the remainder of the broader sample, a matched control participant was identified for each of the participants in the illness group. Matched control participants were selected using the following criteria: (a) he or she was free of significant PH diagnoses through the T5 assessment point, meaning they had not received a significant PH diagnosis in their lives through T5 (or they shared a similar history of mild other illnesses as the participant they were matched with); and (b) he or she exactly matched a participant in the illness group on all of the following demographic categories: sex, age, race, educational attainment. Of these, 17 cases matched identically on all of the matching variables (meaning that 63% of the sample was perfectly matched). For the remainder, the closest possible match was identified (i.e., in some cases the match was 1 year older or younger; in others, the demographic variables were perfectly matched, but the endorsement of "other" previous diagnoses was discordant). In instances for which more than one case provided an equivalent match, the included case was selected at random.

Thus, the complete sample for Study 2 was 54 participants: 27 who received a major illness diagnosis between Baseline and T2 and 27 matched participants who remained healthy throughout the duration of the study. The longitudinal design of Study 2 is presented in Figure 1 for clarity. The results of $t$ tests confirmed that there were no differences between the groups on any demographic characteristic (sex: $t(52) = 0.00$, $p = 1.00$; age: $t(52) = -0.05$, $p = .96$; race: $t(52) = 0.00$, $p = 1.00$; educational attainment: $t(52) = 0.00$, $p = 1.00$), nor were there differences between the groups on MH, $t(52) = -0.01$, $p = .99$, or PH, $t(52) = 0.49$, $p = .51$, at Baseline. The sample was 48% female and 22% African-American/Black. A description of the two samples is included in Table 1. In presenting the results of this study, the two groups will be referred to as the "illness" group and the "control" group. Although some participants had some missing data at some time points, all 54 participants remained in the sample through every assessment. Because the primary longitudinal analytical approach, HLM, can accommodate missing data (e.g., Singer & Willett, 2003), no special procedures were undertaken to account for missing data.

**Measures.**

***MH.*** Participants' MH was assessed at each time point using two measures. First, the BDI-II (Beck et al., 1996; Cronbach's alpha = .81), was used, as in Study 1. In addition, participants' MH was assessed with the Social Adjustment Scale Self-Report (Weissman & Bothwell, 1976; Cronbach's alpha = .82), a 48-item instrument that has been widely used for assessing social functioning in both clinical and community samples, particularly in the context of longitudinal studies of social adjustment. This scale produces both a global score as well as five specific domain scores; in the present study, only the global score was used. Scores on both measures were standardized, and then scores on the BDI-II were reversed such that higher scores on these two measures were both indicative of positive MH. These scores were highly correlated with each other, $r = .54$, $p < .01$; thus, they were standardized and averaged to create a composite score representing participants' MH. Although depression might be seen as a specific outcome domain, given the small sample size, the preliminary nature of this investigation, and the high intercorrelation, a single outcome measure of MH was adopted.

***PH.*** Participants' PH was assessed at each time point with the RAND-36, the most widely used questionnaire measure of health-related quality of life (Hays & Morales, 2001). This instrument is

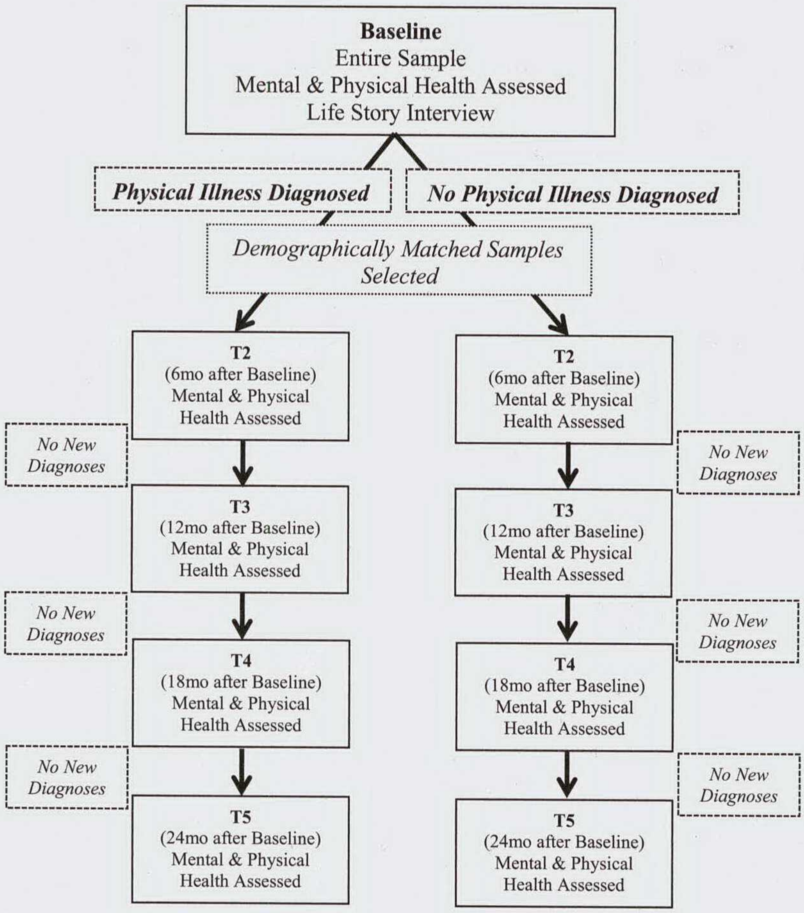


*Figure 1.* Study 2 longitudinal design.

composed of 36 items and produces eight scale scores and two summary scores representing PH and MH. For the purposes of the present study, only the PH composite score from the RAND-36 was used to avoid contamination with the separate assessment of MH. The PH composite includes assessment of physical functioning, pain, and role limitation resulting from PH problems. There are numerous precedents for using the PH composite independently from the rest of the scale as a primary measure of PH, and evidence suggests it is truly distinct from the MH composite (e.g., Simon, Revicki, Grothaus, & Vonkorff, 1998).

***Narratives.*** All participants from the broader data set engaged in an abbreviated Life Story Interview (e.g., McAdams, 2008) at Baseline. Because participants underwent extensive interview-based assessment as part of the broader study from which the present sample was drawn, the Life Story Interview was substantially streamlined to include only the following core scenes: Life Chapters, High Point, Low Point, Turning Point, Positive Characters, and Negative Characters. The narratives were still adequately substantial in length for viable coding ($M$ = 2,883 words, $SD$ = 1,926 words), but much shorter than typical life stories (only 17% the length of the full life stories obtained in Study 1, and 80% the length of the four scenes coded in Study 1). Life Story Interviews were videotaped and professionally transcribed, verbatim.

***Coding of the narratives.*** Participants' entire life story narratives were coded for the same four themes using the same coding systems described in Study 1. Unlike in Study 1, there was no question in the interview that specifically asked participants to narrate a significant PH challenge. Indeed, at Baseline in Study 2, all participants reported being free of the major health conditions that were covered in the diagnostic interview. It seems certain that participants would have been able to generate a health narrative, if directly asked for one, but doing so was beyond the scope of the broader study from which the present data were drawn. In Study 2, two coders (different from those in Study 1), blind to the hypotheses and trained to a high standard of interrater reliability, assessed each life story narrative in its entirety, using the same coding systems used in Study 1, for the themes of agency (ICC Case 2, Type I = 0.80), communion (ICC Case 2, Type I = 0.83), redemption (kappa = 0.86), and contamination (kappa = 0.81).

## Results

**Baseline cross-sectional analyses.** Zero-order correlations presenting the cross-sectional relationships between Baseline demographic, narrative, and MH and PH variables for each group of participants are presented in Table 6. Within both groups, Euro-American/White participants had significantly higher levels of education than African-American/Black participants, and in the control group, higher levels of education were significantly associated with higher income. In addition, for both illness and control

Table 6
*Study 2: Intercorrelations Between Time 1 Variables*

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1. Sex | | | | | | | | | | | | |
| Illness group | | | | | | | | | | | | |
| Control group | | | | | | | | | | | | |
| 2. Race | | | | | | | | | | | | |
| Illness group | −0.20 | | | | | | | | | | | |
| Control group | −0.19 | | | | | | | | | | | |
| 3. Age | | | | | | | | | | | | |
| Illness group | −0.08 | 0.01 | | | | | | | | | | |
| Control group | −0.07 | −0.02 | | | | | | | | | | |
| 4. Income | | | | | | | | | | | | |
| Illness group | −0.16 | 0.29 | −0.13 | | | | | | | | | |
| Control group | −0.20 | 0.20 | 0.22 | | | | | | | | | |
| 5. Education | | | | | | | | | | | | |
| Illness group | −0.16 | 0.67** | −0.12 | 0.32 | | | | | | | | |
| Control group | −0.16 | 0.68** | −0.14 | 0.42* | | | | | | | | |
| 6. Agency | | | | | | | | | | | | |
| Illness group | 0.34 | −0.08 | 0.15 | 0.04 | −0.14 | | | | | | | |
| Control group | 0.04 | 0.11 | 0.03 | 0.20 | 0.30 | | | | | | | |
| 7. Communion | | | | | | | | | | | | |
| Illness group | 0.28 | 0.06 | −0.30 | −0.07 | 0.01 | 0.12 | | | | | | |
| Control group | 0.23 | −0.03 | −0.31 | −0.04 | 0.08 | 0.23 | | | | | | |
| 8. Redemption | | | | | | | | | | | | |
| Illness group | 0.35 | −0.30 | −0.22 | −0.18 | −0.03 | 0.11 | 0.52** | | | | | |
| Control group | 0.36 | 0.22 | −0.28 | −0.12 | 0.07 | 0.30 | 0.52** | | | | | |
| 9. Contamination | | | | | | | | | | | | |
| Illness group | −0.23 | 0.14 | −0.26 | −0.24 | −0.26 | −0.31 | 0.27 | −0.12 | | | | |
| Control group | 0.09 | −0.12 | −0.37* | −0.06 | −0.12 | −0.26 | 0.09 | 0.09 | | | | |
| 10. Word count | | | | | | | | | | | | |
| Illness group | 0.27 | −0.17 | −0.12 | −0.26 | −0.18 | 0.14 | 0.57** | 0.57** | 0.09 | | | |
| Control group | 0.22 | −0.21 | −0.27 | −0.12 | −0.09 | 0.04 | 0.41* | 0.32 | 0.09 | | | |
| 11. Physical health | | | | | | | | | | | | |
| Illness group | 0.24 | 0.35 | −0.22 | 0.20 | 0.11 | 0.11 | 0.29 | 0.04 | −0.01 | 0.15 | | |
| Control group | −0.11 | 0.02 | −0.05 | −0.31 | −0.03 | 0.14 | 0.08 | 0.16 | 0.16 | 0.10 | | |
| 12. Mental health | | | | | | | | | | | | |
| Illness group | −0.08 | 0.49* | −0.02 | 0.09 | 0.31 | 0.29 | 0.18 | 0.30 | −0.10 | −0.22 | 0.14 | |
| Control group | 0.28 | −0.05 | 0.03 | −0.10 | 0.29 | 0.22 | 0.28 | 0.09 | −0.09 | −0.23 | 0.15 | |

* $p < .05$. ** $p < .01$.

groups, the themes of redemption and communion were significantly positively correlated with each other. Furthermore, the length of narratives was significantly associated with the themes of redemption and contamination. Previous research has not reported an association between word count and these themes.

Across groups, there were no significant associations between the narrative themes and either MH or PH in concurrent analyses at T1, though all trends were in the expected directions. This null funding, specifically with respect to MH, was unexpected and counter to our hypotheses. It may be an artifact of the relatively small sample size of this study and/or the comparatively small amount of narrative data available for coding. Indeed, moderate correlations were observed between MH and the themes of agency, communion, and redemption, even though they did not reach standard significance levels.

In preparation for longitudinal analyses comparing participants in the illness and control groups, it was necessary to determine whether there were any significant differences between the narratives in each group. When the narrative themes were assessed across the two groups, there were no significant differences at Baseline between the illness group and the control group (agency: $t(52) = 0.90$, $p = .37$; communion: $t(52) = -0.23$, $p = .82$; redemption: $t(52) = 1.15$, $p = .25$; contamination: $t(52) = 0.20$, $p = .85$). There was also no significant difference in the mean length of narratives written by the two groups of participants, $t(52) = -0.66$, $p = .52$. When this result is considered along with the matched demographic variables and nonsignificant differences in MH or PH, it is clear that there are no significant differences measured between the participants in the illness and control groups prior to one group being diagnosed with an illness.

**Longitudinal analyses.**

***Analyses across the entire sample.*** The first set of growth curve models was constructed to test whether there were significant differences in the trajectories of MH and PH between the illness and control groups. Longitudinal models for Study 2 are presented in Table 7. Not surprisingly, when group (illness vs. control) was entered as a Level 2 variable, significant differences in PH were observed over time between the illness participants and the control participants (Model 1). This affirms that the diagnosis of an illness between Baseline and T2 in the illness group did result in poorer trajectories of PH in this group compared with the control group. No significant differences were observed in MH over time between the two groups (Model 2). Following these analyses, the data were split into two groups based on participants'

Table 7
*Study 2 Growth Curve Models (Final Estimation of Fixed Effects With Robust Standard Errors)*

| Parameter | Coefficient | *SE* | *t* |
|---|---|---|---|
| Model 1: Physical health over time with group at Level 2 | | | |
| Intercept | 55.15 | 1.82 | 30.27*** |
| Slope | 0.81 | 0.40 | 2.07* |
| Group | 6.04 | 1.96 | 3.08** |
| Model 2: Mental health over time with group at Level 2 | | | |
| Intercept | 0.03 | 0.18 | 0.17 |
| Slope | −0.02 | 0.04 | −0.64 |
| Group | 0.08 | 0.21 | 0.35 |
| Model 3: Physical health over time within illness group | | | |
| Intercept | 55.47 | 1.70 | 34.37** |
| Slope | −0.76 | 0.35 | −2.16* |
| Model 4: Physical health over time within illness group with illness type at Level 2 | | | |
| Intercept | 52.72 | 3.64 | 14.49** |
| Slope | −0.58 | 0.68 | −0.85 |
| Illness type | 0.17 | 0.96 | 0.19 |
| Model 5: Mental health over time within illness group | | | |
| Intercept | 0.07 | 0.10 | 0.46 |
| Slope | −0.03 | 0.04 | −0.65 |
| Model 6: Mental health over time within illness group with narrative themes at Level 2 | | | |
| Intercept | 50.69 | 3.49 | 14.54*** |
| Slope | 0.79 | 0.34 | 2.36* |
| Agency | 4.30 | 1.17 | 2.13* |
| Communion | 6.16 | 2.25 | 2.74* |
| Redemption | 14.56 | 6.45 | 2.26* |
| Contamination | −9.72 | 6.72 | −1.95* |
| Model 7: Relationship between mental health and physical health over time within illness group with narrative themes at Level 2 | | | |
| Intercept | 49.59 | 5.09 | 9.75*** |
| Mental health slope | 0.99 | 0.48 | 2.04* |
| Physical health slope | 0.01 | 0.07 | 0.19 |
| Agency | 3.46 | 1.95 | 1.98* |
| Communion | 6.38 | 2.70 | 2.36* |
| Redemption | 16.47 | 7.91 | 2.22* |
| Contamination | −10.49 | 7.94 | −1.72* |
| Model 8: Mental health over time within control group | | | |
| Intercept | 62.68 | 1.84 | 34.00** |
| Slope | −0.41 | 0.46 | −0.88 |
| Model 9: Physical health over time within control group | | | |
| Intercept | 58.02 | 1.21 | 47.95** |
| Slope | 0.17 | 0.28 | 0.61 |
| Model 10: Mental health over time within control group with narrative themes at Level 2 | | | |
| Intercept | 69.86 | 3.49 | 20.00*** |
| Slope | 0.41 | 0.46 | 0.89 |
| Agency | 3.30 | 2.64 | 1.25 |
| Communion | 1.46 | 2.41 | 0.60 |
| Redemption | 13.12 | 12.00 | 1.09 |
| Contamination | −20.68 | 7.29 | −2.84* |
| Model 11: Physical health over time within control group with narrative themes at Level 2 | | | |
| Intercept | 49.27 | 7.62 | 6.47*** |
| Slope | 0.22 | 0.43 | 0.52 |
| Agency | 2.22 | 3.43 | 0.65 |
| Communion | −1.20 | 4.34 | −0.28 |
| Redemption | 14.10 | 16.98 | 0.83 |
| Contamination | 16.82 | 12.85 | 1.31 |

$^{*}p < .05$. $^{**}p < .01$. $^{***}p < .001$.

diagnostic status, and subsequent models were run separately for each group.

***Analyses within the illness group.*** Within the illness group, significant decline in PH was observed over time (Model 3). Within this group, there were no significant differences in PH decline over time across the six diagnostic categories that were assessed (diagnostic type was entered as a Level 2 variable; Model 4). In other words, participants in this group reported comparable levels of PH decline on the RAND-36, regardless of the specific diagnosis they received. Within this group, there was no significant decline in MH over time (Model 5) and no significant variability in MH changes across diagnostic groups. In addition, there were no differences in either MH or PH over time across participants' sex, age, race, education level, or personal income.

Although illness participants experienced declines in their PH, but not MH, over time, a different pattern of results emerged when variability in participants' baseline narratives was taken into account. A model that examined the relationship between individual differences in the four narrative themes and intraindividual trajectories of MH over time in this group indicated that participants whose narratives included higher levels of the themes of agency, communion, and redemption, and lower levels of the theme of contamination, experienced positive trajectories in their MH compared with participants with contrasting themes (Model 6). In other words, among participants who were diagnosed with a significant illness, individual differences in the thematic content of their life stories prior to their diagnosis were associated with intraindividual trajectories of MH following diagnosis.

In addition, the variability in participants' trajectories of MH over time based on the themes in their narratives was not significantly impacted by including their trajectory of PH over time in the model (Model 7). This suggests that the themes of agency, communion, redemption, and contamination were associated with participants' MH over time, regardless of their PH over time.

Following the results of exploratory analyses from Study 1, a parallel set of exploratory analyses was conducted, examining the associations between individual differences in the narratives and trajectories of MH over time, using narrative codes from each scene independently. There were no significant results of these exploratory analyses, suggesting that there was no specificity to scene-level analyses in this sample. This is likely related to the very small sample size and the short length of each narrative scene.

***Analyses within the control group.*** Within the control group, no significant changes in MH or PH were observed over time (Models 8 and 9). In addition, there were no differences in either MH or PH over time across participants' sex, race, education level, or personal income. There was, however, a marginally significant impact of participant age on the change in MH over time, such that older participants had marginally better MH over time (age: $\beta_{01} = 1.39$, $t(24) = 2.53$, $p = .06$).

For the most part, the narratives of participants in the control group were not significantly associated with their MH or PH over time (Models 10 and 11). However, the theme of contamination was significantly associated with negative trajectories of MH over time. Exploratory analyses yielded no significant associations between narrative themes and MH over time when analyzed by scene.

## Illustrative Case Example

Given the small sample size included in the primary within-group analyses, a brief case example both illustrates the central findings and grounds them in the rich experiences described by each participant in this study. In narrative research, participants are regarded as the experts on their own lives; their experience of participating in a study ought to provide them with an opportunity to share their unique approach to meaning-making in a way that is ultimately instructive (e.g., Josselson & Lieblich, 1993). Especially in the context of a small-sample study, such as Study 2, it is important to return to participants' voices as the site where quantitative trends are instantiated in individual lives.

Ava[1] grew up in rural southern Illinois and moved to Alton, Missouri, when she got married at the age of 19. She remembers the date in November of 1967 when she got engaged and the day she moved to Alton in May of 1968, the first time she lived away from her parents. Unfortunately, the joy of that heady time did not last. Within 5 years, she had two children and her marriage had started to go downhill. In recounting her life story, she described enduring years of mental abuse and occasional physical abuse at the hands of her husband. But in spite of this, she stayed married to him for 26 years. In the early years, she was happy to be a stay-at-home mother to her children, but later, when she contemplated divorce, she realized that this arrangement had left her without any job history, and she worried about the prospects of earning enough money to raise two children on her own. She worried more about being ostracized from her community, the only community she had known since her childhood, because her husband was an elder in their church and well-respected in town. Ava's marriage did not end with a climactic moment in which she fled the house, fearing for her life. The ending began gradually, during a typical conversation.

One winter in the early 1990s, Ava and her husband took a vacation in Florida. They drove from Alton to the Gulf Coast to visit her brother-in-law and spend some time in the sun. She described it this way:

> We were walking on the beach and, I forget exactly where it was, but anyhow, he said, "I love you," and I didn't respond. And, he said, "So, you don't love me anymore?" And, I said, "I, I respect you, but I don't love you." And then, um, we had a long drive home, back to Alton. And then probably, I'd say within a year, I left.

Ava singles out the day she left her husband as the turning point of her life:

> As I pulled out of the driveway, I have to admit, I felt like a bird out of the gilded cage. I felt like I had just been released from all sorts of, um, abuse that I had been suffering and that type of thing. And, it was the beginning of a new life . . . So, I went ahead and moved without any furniture. I kid you not. No furniture at all. I slept on the floor; I made a pallet on the floor in the bedroom. I had no TV, no kitchen table, um, no couch, no chair. And that was it. But, I knew that I was going to start from scratch on the ground, going on my way up.

She got an apartment in St. Louis, got a divorce attorney with the help of her parents, and took out a protective order against her husband. This new chapter of her life was quite complicated. Her daughter was not happy with Ava's decision to involve the police, and Ava was adjusting to being a working woman in a big city. She continued,

> So, then, um, for a period of time my life just was going to work and coming back home, going to work and coming back home. And, then I went ahead and, um, met a gentleman at work and helped him out

[1] Certain potentially identifying details, including names, have been modified in this case example to protect confidentiality.

> with a couple things (find an apartment and stuff like this) and, uh, that gentleman is now my husband.

Ava did not rush in to marrying the man she had met at work; they dated for 10 years before tying the knot.

Her second husband, Robert, was a bus driver for the city, a job he loved. Ava said,

> I call him, lovingly, "Mr. Billy Joe Friendly Ray Bob," because he meets and greets his passengers, just loves them all. Hands out little stuffed animals to the kids that get on the bus and everything. And, um, goes out of his way to get people to their destination safely.

He had also been previously married, but never had children, and over time Ava's children and grandchildren came to see Robert as part of the family, despite some of their initial resistance to her marrying an African American man (a tension that was diffused by a humorous incident when Ava's 5-year-old grandson tried to wipe the "dirt" off of Robert's face).

In the few years prior to Ava's participation in the study, Robert contracted a virulent bacterial infection that nearly killed him and left him unable to work, but even after an extremely rare reinfection in the months just prior to the interview that required one of Robert's legs to be amputated, Ava described their marriage as a sanctuary: "He's at home and he's mending well . . . It's just the two of us and everything's really good."

At the very end of the life story interview, Ava looked back over her life and compared the immensely positive role Robert had played in her life with that of her first husband. She returned to the years of abuse she endured and singled out her ex as the most negative character in her life story. She ended the interview by saying,

> But, I can turn it around also and make it a positive, because through his actions and his behaviors, I learned how to literally save myself, get out of a bad situation and, go on with my life. Because I don't know what my life would be like today if I stayed over there.

This final paragraph echoes the themes that ran throughout Ava's life story—the themes of agency and redemption. Here, she saw how the years of suffering taught her how to take charge of her own life—a theme that is also found in the story of the day she left her husband, when she "[started] from scratch on the ground, going on [her] way up." Her reference to feeling "like a bird out of the gilded cage" is beautifully evocative of redemption through liberation—one of the four major variations on redemption in American cultural narratives, as described by McAdams (2013). In both the story of leaving her first husband and her life reflection, Ava narrated an extremely negative experience as being transformed into a positive outcome. And, in both instances, she also described how it was her own efforts that were ultimately responsible for the positive change. Ava's story is also rich in communal themes, especially in the stories of connection with Robert, and nearly free from contamination sequences. Indeed, Ava's life story scored high in agency, communion, and redemption, and low in contamination.

At the second assessment point, 6 months after completing the life story interview, Ava reported that she had been diagnosed with breast cancer in the intervening period. Her MH, which had been just about at the mean of the overall sample at Baseline (a $z$ score of 0.12), took a dip (to 0.04), though not a major one, like her PH score (a drop from a $z$ score of 0.15 to $-1.77$). Figure 2 displays the trajectories of Ava's mental and PH scores over time. But by 6 months later, 1 year after the life story interview, the trajectory of her MH turned upward and continued rising, even though her PH scores rose and then flattened out.

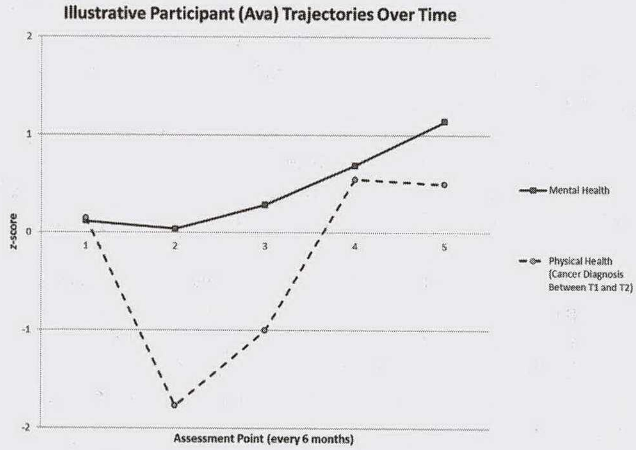


*Figure 2.* Illustrative case study. Mental health composite and physical health composite (cancer diagnosis between Time 1 and Time 2).

Without an additional life story interview, it is impossible to know how Ava made sense of her cancer diagnosis. But the general trends observed in the group of participants in this study who fell ill between Baseline and T2, suggest that the way Ava narrated her life may have served her well following this new challenge. Indeed, being diagnosed with cancer was not the first hardship in Ava's life, and she developed a narrative approach to making meaning of her difficult experiences that featured her own agency and her ability to draw redemptive lessons (and, to a certain extent, her connections with others). As such, her narrative identity may be supporting her MH in the wake of this misfortune. This case illustrates the ways in which particular approaches to meaning-making in the life story may serve a protective function for people when they encounter new adversity.

## Discussion

The results of Study 2 drew on a specially selected sample to examine the associations between individual differences in narrative identity and intraindividual trajectories of MH over the course of 2 years in order to examine the impact of a specific negative life event: receiving a major physical illness diagnosis.

At Baseline, when all participants in the sample reported no major physical illnesses, there were no meaningful differences measured between groups: They were selected for matched demographics, participants in both groups reported similar PH and MH, and there were no significant differences in the narrative themes across the groups. Somewhat surprisingly, there were no significant associations between the narrative themes and MH in either group when assessed concurrently, though all associations were in the predicted directions, and correlations were modest in size, despite not reaching conventional significance levels. This result is likely an artifact of the relatively small sample size and the comparatively short narratives in Study 2. Nevertheless, when individual differences in participants' narratives were compared with their trajectories of MH over the following 2 years, significant relationships emerged among those participants who experienced the onset of a major physical illness.

Within this group, high levels of the themes of agency, communion, and redemption, and low levels of the theme of contamination, were associated with positive trajectories of MH following the diagnosis. Furthermore, the association between variability in narrative themes and MH over time was not impacted by the trajectory of their PH over time. In other words, the themes in participants' life stories, obtained prior to becoming ill, were associated with the trajectory of their MH following diagnosis, regardless of their PH. No such pattern of associations was observed in participants who remained healthy throughout the 2 years of the study.

This central result suggests that particular ways of narrating one's life may play a protective role for MH in the wake of new, difficult experiences. It aligns with the conclusions from a recent study that found that among participants who had experienced recent "biographical disruptions" (events which instilled a sense of self-discontinuity), autobiographical reasoning coded in their narratives of key life events was associated with repaired self-continuity (Habermas & Köber, 2014). An analogous effect was not observed in participants who did not experience recent biographical disruptions. Together, these findings may suggest that narrative identity exerts its effects on MH most strongly when new life events challenge one's existing self-story. The case of Ava grounds these general trends in the specific life experiences of a representative participant.

## General Discussion

The pair of studies presented here represents the first evidence that individual differences in narrative identity are associated with trajectories of MH over substantial periods of time. In two studies drawing on samples of late-mid-life adults, variation in the themes of agency, redemption, and contamination (and, in the second of the two studies, communion) were associated with different trajectories of intraindividual change in MH over several years. In Study 1, higher levels of agency and redemption, and lower levels of contamination, in core sections of participants' life stories were associated with increases in MH over 4 years. In Study 2, higher levels of agency, communion, and redemption, and lower levels of contamination, in participants' life stories were associated with increases in MH over 2 years among participants who were diagnosed with a major physical illness in the 6 months after their life stories were collected. Taken together, these results suggest that particular approaches to narrating one's life experiences may support MH over time and may play an especially potent protective role in the wake of new challenges. These results substantially extend prior work that has documented cross-sectional associations between these narrative themes and MH and studies examining change in MH using two-time-point designs. Specifically, the present studies are the first to undertake a direct investigation of the association between individual differences in narrative identity and intraindividual trajectories of change in MH over time.

The two studies focused on the developmental period of late midlife as an opportunity for examining its core hypotheses during a time when one particular challenge is especially salient, the decline in PH common during this period. The relationship between the body and identity is salient throughout the life course. Erikson (1959) suggested that the biological changes of puberty are one of the key factors in spurring the emergence of identity in adolescence. Illness in adulthood often serves as an autobiographical disruption (e.g., Bury, 1982). In addition to the physical pain and the financial burden that come with being sick, the shift from person to patient may bring with it feelings of disempowerment and isolation. Making sense of these changes is a challenge to both MH and narrative identity (e.g., Bury, 1982; Kenyon et al., 2011). In both studies, variation in narrative identity was associated with participants' MH, regardless of the status of their PH. This finding is especially notable in Study 2, in which the significant associations between variability in narrative themes and trajectories of MH were not significantly impacted by the inclusion of PH, even among the group of participants who were diagnosed with a major physical illness during the course of the study. This suggests that the associations between the subjective meaning made using the themes of agency, communion (in Study 2), redemption, and contamination and participants' MH over time was not related to their actual PH.

The results from exploratory analyses in Study 1 also extend previous research on the connection between narrative identity and MH. In this study, the themes of agency, redemption, and contamination in stories of low points and health challenges were associated with MH, whereas these themes in stories of high points and turning points were not. Although scene-level analyses in Study 2 did not replicate these results, another set of findings does compliment them. In Study 2, the themes of agency, communion, redemption, and contamination were associated with trajectories of MH among those participants diagnosed with a major illness, but not among those who remained healthy. When considered alongside the findings from exploratory analyses in Study 1, the broader collection of results suggests that narrative identity may be especially relevant to MH in the context of adults' life challenges. In Study 1, narratives of hardship (health challenges and low points) seemed to drive the overall pattern of results connecting variability in narrative themes to MH. In Study 2, the connections between variability in narrative themes and MH were observed among those participants who experienced a substantially difficult life experience. Although these two sets of findings are not perfectly parallel, taken together, they do open up a potentially fruitful avenue for further exploration—examining whether negative life events are the key locus for the association between narrative identity and MH in adulthood, both when evaluated as the content of life stories and as the context in which life stories exert the strongest influence on MH (see also Habermas & Köber, 2014).

The naturalistic, nonexperimental design of these two studies does not allow for a direct answer to the question of why the themes of agency, communion, redemption, and contamination might be especially strongly associated with MH. These results do suggest that these four themes in personal narratives may hold special potency for contemporary American adults in the face of personal challenges; nevertheless, the specific mechanism of this association remains elusive. It is possible that constructing a personal narrative that adopts these ubiquitous master themes from American culture (e.g., Hammack, 2008; McAdams, 2006) provides a psychological sense of connection with the broader society, or a sense that a given person is not isolated in his or her individual struggles—that his or her stories sound the way life stories ought to sound. It is possible that using these themes enhances an individual's sense that his or her fundamental needs for autonomy and relatedness (e.g., Deci & Ryan, 2000; Philippe et al., 2011) are being fulfilled, despite life's challenges. Future research designed specifically to examine the mechanism of connection between these themes and MH is certainly warranted.

Although this pair of studies suggests that there may be an especially strong association between narrative identity and MH in the context of major life challenges, such as physical illness, it is vital to note that no one hopes for a PH challenge as an opportunity for psychological growth. The positive results of these two studies concerning the themes of agency, communion, redemption, and contamination should not be interpreted to suggest that the potential benefits of these experiences outweigh the suffering they inevitably bring. PH challenges are among the most frightening, most existentially traumatizing experiences people undergo. It is hopeful that different narrative choices may powerfully impact one's MH surrounding these experiences, but narrative identity is certainly not a panacea, even in the purely subjective realm of meaning-making. It is also vital that the results of these studies not be interpreted to suggest that people who are ill and who have not crafted agentic or redemptive narratives are to be regarded as weaker or deficient in any way. In the face of illness, agency and redemption (and communion) are not a given. They may be powerful tools, evidence of courage, and a psychological accomplishment in their own right, but their absence in a story of sickness is not itself a failure.

It is also important to remember that, regardless of the specific themes being assessed, narratives are not optimal tools for evaluating the objective reality of an individual's situation. Instead, narratives provide an ideal vehicle for assessing the subjective meaning-making that individuals bring to their experiences. The correspondence between the subjective meaning and the objective reality may be more or less aligned, but as research on narratives of life's challenges from a wide variety of contexts makes clear, the thematic components of one's narrative may impact one's MH in the face of adversity. Thus, whether the narrative of the agentic fighter is more myth than reality, for example, is not actually the central point when it comes to impacting MH. Instead, the story itself may support and foster positive MH, regardless of the actual circumstances. In addition, a study that examined questionnaire-based assessment of agency and communion in a sample of adolescents with Type 1 diabetes and a matched sample of healthy adolescents found no differential associations were observed with regard to these constructs and participants' MH across groups (Helgeson & Palladino, 2012). Nevertheless, exploring the connections between narrative identity and PH would be an interesting avenue for future investigation.

Given the naturalistic setting of both studies, there are certain limitations to the conclusions that can be drawn from these results. Most significantly, in neither study was PH assessed by a physician, which may limit the validity of the diagnoses reported by participants. Although there were no clear incentives for participants to lie about their PH, it is possible that participants were misinformed, incorrect, or forgetful about their health status. Nevertheless, PH was assessed via widely used instruments developed for assessing self-reported PH, each of which has a very strong tradition of research in the medical literature (e.g., Hays & Morales, 2001; Ware, Kosinski, & Keller, 1994). It is important to note that we did not focus on fine-grained differentiation of different medical diagnoses in the present study, and when broad distinctions were included in analyses (in Study 2), the results were similar across a wide range of illness categories.

A second limitation to the present pair of studies is that narratives themselves were not collected at each time point in the investigations. This meant that it was not possible to examine the longitudinal trajectories of the narrative themes alongside the trajectories of MH and PH, as has been done in the past (e.g., Adler, 2012; Adler & Hershfield, 2012). As such, it is also not possible to determine whether the mechanism of these associations is a result of the active construction of the life story that occurred as part of participation or was derived from preexisting personality variables that were expressed in these life stories. Although the particular prospective models reported in the present pair of studies break new analytic ground in the study of narratives, analyses of this sort would have provided a more dynamic evaluation of the relationship between personal narratives and mental and PH over time.

Third, Study 2, in particular, should be interpreted as quite preliminary, given the small and unusual sample and the possibility that other, nonassessed negative experiences may have been shared across the two groups. The opportunity to examine the association between narrative identity and trajectories of MH in a highly controlled set of circumstances is unprecedented, so the results of this study break new empirical ground. Nevertheless, the winnowing down of a much larger, epidemiologically representative sample to this small, select group who met very specific criteria certainly introduces the likelihood that unmeasured third variables may impact the conclusions. Indeed, the failure of this study to replicate certain commonly identified cross-sectional findings should serve as evidence of its unusualness in the narrative literature. Despite these limitations, the methodological and analytical approaches in this study do permit the investigation of certain questions that cannot be addressed using approaches adopted in prior research, and, as such, the results should preliminarily add to and extend our understanding of the connections between narrative identity and MH.

Finally, although the samples in each study represent a diverse group of adults, the age range of participants in both studies was restricted to people in their 50s and 60s. The relationship between personal narratives and MH in the context of declining PH is especially relevant for late-mid-life adults, but it is unclear whether the findings from this pair of studies would generalize to individuals at different periods in the life course.

## Conclusion

The present pair of studies suggests that variability in narrative identity is not only associated with MH concurrently but is also associated with trajectories of intraindividual variation in MH over time. Drawing on samples of late-mid-life adults followed over the course of several years, it extends previous research that has documented the associations between the themes of agency, communion, redemption, and contamination and MH, demonstrating that these themes may also exert prospective influences. These associations may be especially strong when concerned with negative events: the themes in stories of challenging circumstances appeared to drive the overall findings in Study 1, and the themes were most strongly associated with MH among participants who experienced a new life challenge in Study 2. Taken together, this pair of studies should both bolster theoretical assertions and existing empirical evidence about the relationship between narrative identity and MH, and should encourage future work examining these associations over time. It also suggests that narrative identity may play a central role in healthy aging.

## References


Adler, J. M. (2012). Living into the story: Agency and coherence in a longitudinal study of narrative identity development and mental health over the course of psychotherapy. *Journal of Personality and Social Psychology, 102,* 367–389. http://dx.doi.org/10.1037/a0025289

Adler, J. M., Chin, E. D., Kolisetty, A. P., & Oltmanns, T. F. (2012). The distinguishing characteristics of narrative identity in adults with features of borderline personality disorder: An empirical investigation. *Journal of Personality Disorders, 26,* 498–512. http://dx.doi.org/10.1521/pedi.2012.26.4.498

Adler, J. M., & Hershfield, H. E. (2012). Mixed emotional experience is associated with and precedes improvements in psychological well-being. *PLoS ONE, 7*(4), e35633. http://dx.doi.org/10.1371/journal.pone.0035633

Adler, J. M., Kissel, E. C., & McAdams, D. P. (2006). Emerging from the CAVE: Attributional style and the narrative study of identity in midlife adults. *Cognitive Therapy and Research, 30,* 39–51. http://dx.doi.org/10.1007/s10608-006-9005-1

Adler, J. M., Lodi-Smith, J., Philippe, F. L., & Houle, I. (in press). The incremental validity of narrative identity in predicting well-being: A review of the field and recommendations for the future. *Personality and Social Psychology Review.*

Adler, J. M., & Poulin, M. J. (2009). The political is personal: Narrating 9/11 and psychological well-being. *Journal of Personality, 77,* 903–932. http://dx.doi.org/10.1111/j.1467-6494.2009.00569.x

Adler, J. M., Skalina, L. M., & McAdams, D. P. (2008). The narrative reconstruction of psychotherapy and psychological health. *Psychotherapy Research, 18,* 719–734. http://dx.doi.org/10.1080/10503300802326020

Aspinwall, L. G., & Tedeschi, R. G. (2010). Of babies and bathwater: A reply to Coyne and Tennen's views on positive psychology and health. *Annals of Behavioral Medicine, 39,* 27–34. http://dx.doi.org/10.1007/s12160-010-9155-y

Baddeley, J., & Singer, J. A. (2008). Telling losses: Personality Correlates and Functions of bereavement narratives. *Journal of Research in Personality, 42,* 421–438.

Baddeley, J., & Singer, J. A. (2010). A loss in the family: Silence, memory, and narrative identity after bereavement. *Memory (Hove, England), 18,* 198–207. http://dx.doi.org/10.1080/09658210903143858

Bandura, A. (2006). Toward a psychology of human agency. *Perspectives on Psychological Science, 1,* 164–180. http://dx.doi.org/10.1111/j.1745-6916.2006.00011.x

Bauer, J. J., & McAdams, D. P. (2004). Personal growth in adults' stories of life transitions. *Journal of Personality, 72,* 573–602. http://dx.doi.org/10.1111/j.0022-3506.2004.00273.x

Bauer, J. J., & McAdams, D. P. (2010). Eudaimonic growth: Narrative growth goals predict increases in ego development and subjective well-being 3 years later. *Developmental Psychology, 46,* 761–772. http://dx.doi.org/10.1037/a0019654

Bauer, J. J., McAdams, D. P., & Sakaeda, A. R. (2005). Interpreting the good life: Growth memories in the lives of mature, happy people. *Journal of Personality and Social Psychology, 88,* 203–217. http://dx.doi.org/10.1037/0022-3514.88.1.203

Beck, A. T., Steer, R. A., & Brown, G. K. (1996). *Manual for Beck Depression Inventory.* San Antonio, TX: Psychological Corporation.

Berntsen, D., & Rubin, D. C. (2004). Cultural life scripts structure recall from autobiographical memory. *Memory & Cognition, 32,* 427–442. http://dx.doi.org/10.3758/BF03195836

Bohn, A. (2011). Normative ideas of life and autobiographical reasoning in life narratives. *New Directions for Child and Adolescent Development, 2011*(131), 19–30. http://dx.doi.org/10.1002/cd.286

Bryk, A. S., & Raudenbush, S. W. (1987). Application of hierarchical linear models to assessing change. *Psychological Bulletin, 101,* 147–158. http://dx.doi.org/10.1037/0033-2909.101.1.147

Bury, M. (1982). Chronic illness as biographical disruption. *Sociology of Health & Illness, 4,* 167–182. http://dx.doi.org/10.1111/1467-9566.ep11339939

Chipperfield, J. G., Newall, N. E., Perry, R. P., Stewart, T. L., Bailis, D. S., & Ruthig, J. C. (2012). Sense of control in late life: Health and survival implications. *Personality and Social Psychology Bulletin, 38,* 1081–1092. http://dx.doi.org/10.1177/0146167212444758

Christakis, N. A. (2004). Social networks and collateral health effects. *BMJ: British Medical Journal, 329,* 184–185. http://dx.doi.org/10.1136/bmj.329.7459.184

Collins, L. M., & Sawyer, A. G. (2000). Modeling growth and change processes: Design, measurement, and analysis for research in social psychology. In H. T. Reis & C. M. Judd (Eds.), *Handbook of research methods in social and personality psychology* (pp. 478–495). New York, NY: Cambridge University Press.

Coyne, J. C., & Tennen, H. (2010). Positive psychology in cancer care: Bad science, exaggerated claims, and unproven medicine. *Annals of Behavioral Medicine, 39,* 16–26. http://dx.doi.org/10.1007/s12160-009-9154-z

Deci, E. L., & Ryan, R. M. (2000). The "what" and "why" of goal pursuits: Human needs and the self-determination of behavior. *Psychological Inquiry, 11,* 227–268. http://dx.doi.org/10.1207/S15327965PLI1104_01

Diener, E., Emmons, R. A., Larsen, R. J., & Griffin, S. (1985). The Satisfaction With Life Scale. *Journal of Personality Assessment, 49,* 71–75. http://dx.doi.org/10.1207/s15327752jpa4901_13

Dunlop, W. L., & Tracy, J. L. (2013). Sobering stories: Narratives of self-redemption predict behavioral change and improved health among recovering alcoholics. *Journal of Personality and Social Psychology, 104,* 576–590. http://dx.doi.org/10.1037/a0031185

Erikson, E. H. (1959). *Identity and the life cycle.* New York, NY: Norton.

Garson, G. D. (2012). *Hierarchical linear modeling: Guide and applications.* New York, NY: Sage.

Giddens, A. (1991). *Modernity and self-identity: Self and society in the late-modern age.* Stanford, CA: Stanford University Press.

Graybeal, A., Sexton, J. D., & Pennebaker, J. W. (2002). The role of story-making in disclosure writing: The psychometrics of narrative. *Psychology & Health, 17,* 571–581. http://dx.doi.org/10.1080/08870440290025786

Habermas, T., & Bluck, S. (2000). Getting a life: The emergence of the life story in adolescence. *Psychological Bulletin, 126,* 748–769. http://dx.doi.org/10.1037/0033-2909.126.5.748

Habermas, T., & Köber, C. (2014). Autobiographical reasoning in life narratives buffers the effect of biographical disruptions on the sense of self-continuity. *Memory (Hove, England),* 1–11. http://dx.doi.org/10.1080/09658211.2014.920885

Hammack, P. L. (2008). Narrative and the cultural psychology of identity. *Personality and Social Psychology Review, 12,* 222–247. http://dx.doi.org/10.1177/1088868308316892

Hays, R. D., & Morales, L. S. (2001). The RAND-36 measure of health-related quality of life. *Annals of Medicine, 33,* 350–357. http://dx.doi.org/10.3109/07853890109002089

Helgeson, V. S. (1994). Relation of agency and communion to well-being: Evidence and potential explanations. *Psychological Bulletin, 116,* 412–428. http://dx.doi.org/10.1037/0033-2909.116.3.412

Helgeson, V. S., & Palladino, D. K. (2012). Agentic and communal traits and health: Adolescents with and without diabetes. *Personality and Social Psychology Bulletin, 38,* 415–428. http://dx.doi.org/10.1177/0146167211427149

Johnson, T. (1998). Medicine and the media. *The New England Journal of Medicine, 339,* 87–92. http://dx.doi.org/10.1056/NEJM199807093390206

Jones, B. K., & McAdams, D. P. (2013). Becoming generative: Socializing influences recalled in life stories in late midlife. *Journal of Adult Development, 20,* 158–172. http://dx.doi.org/10.1007/s10804-013-9168-4

Josselson, R., & Lieblich, A. (1993). *The narrative study of lives*. Thousand Oaks, CA: Sage.

Kenyon, G., Bohlmeijer, E., & Randall, W. E. (2011). *Storying later life: Issues, investigations, and interventions in narrative gerontology*. New York, NY: Oxford University Press.

Keyes, C. L. M. (1998). Social well-being. *Social Psychology Quarterly, 61,* 121–140. http://dx.doi.org/10.2307/2787065

King, L. A., & Raspin, C. (2004). Lost and found possible selves, subjective well-being, and ego development in divorced women. *Journal of Personality, 72,* 603–632. http://dx.doi.org/10.1111/j.0022-3506.2004.00274.x

King, L. A., Scollon, C. K., Ramsey, C., & Williams, T. (2000). Stories of life transition: Subjective well-being and ego development in parents of children with Down syndrome. *Journal of Research in Personality, 34,* 509–536. http://dx.doi.org/10.1006/jrpe.2000.2285

King, L. A., & Smith, N. G. (2004). Gay and straight possible selves: Goals, identity, subjective well-being, and personality development. *Journal of Personality, 72,* 967–994. http://dx.doi.org/10.1111/j.0022-3506.2004.00287.x

Kleinman, A., & Seeman, D. (1999). Personal experience of illness. In G. L. Albrecht, R. Fitzpatrick, & S. C. Scrimshaw (Eds.), *The handbook of social studies in health and medicine* (pp. 230–242). New York, NY: Sage.

Laurenceau, J.-P., Hayes, A. M., & Feldman, G. C. (2007). Some methodological and statistical issues in the study of change processes in psychotherapy. *Clinical Psychology Review, 27,* 682–695. http://dx.doi.org/10.1016/j.cpr.2007.01.007

Lodi-Smith, J., Geise, A. C., Roberts, B. W., & Robins, R. W. (2009). Narrating personality change. *Journal of Personality and Social Psychology, 96,* 679–689. http://dx.doi.org/10.1037/a0014611

Lysaker, P. H., Erikson, M., Macapagal, K. R., Tunze, C., Gilmore, E., & Ringer, J. M. (2012). Development of personal narratives as a mediator of the impact of deficits in social cognition and social withdrawal on negative symptoms in schizophrenia. *Journal of Nervous and Mental Disease, 200,* 290–295. http://dx.doi.org/10.1097/NMD.0b013e31824cb0f4

Mansfield, C., McLean, K. C., & Lilgendahl, J. P. (2010). Narrating traumas and transgressions: Links between narrative processing, wisdom, and well-being. *Narrative Inquiry, 20,* 246–273. http://dx.doi.org/10.1075/ni.20.2.02man

Mansfield, E. D., & McAdams, D. P. (1996). Generativity and themes of agency and communion in adult autobiography. *Personality and Social Psychology Bulletin, 22,* 721–731. http://dx.doi.org/10.1177/0146167296227006

McAdams, D. P. (1995). What do we know when we know a person? *Journal of Personality, 63,* 365–396. http://dx.doi.org/10.1111/j.1467-6494.1995.tb00500.x

McAdams, D. P. (2001). The psychology of life stories. *Review of General Psychology, 5,* 100–122. http://dx.doi.org/10.1037/1089-2680.5.2.100

McAdams, D. P. (2006). The problem of narrative coherence. *Journal of Constructivist Psychology, 19,* 109–125. http://dx.doi.org/10.1080/10720530500508720

McAdams, D. P. (2008). *The Life Story Interview*. Retrieved from http://www.sesp.northwestern.edu/foley/instruments/interview/

McAdams, D. P. (2013). *The redemptive self: Stories Americans live by* (rev. ed.). New York, NY: Oxford University Press.

McAdams, D. P., Bauer, J. J., Sakaeda, A. R., Anyidoho, N. A., Machado, M. A., Magrino-Failla, K., . . . Pals, J. L. (2006). Continuity and change in the life story: A longitudinal study of autobiographical memories in emerging adulthood. *Journal of Personality, 74,* 1371–1400. http://dx.doi.org/10.1111/j.1467-6494.2006.00412.x

McAdams, D. P., Hoffman, B. J., Day, R., & Mansfield, E. D. (1996). Themes of agency and communion in significant autobiographical scenes. *Journal of Personality, 64,* 339–377. http://dx.doi.org/10.1111/j.1467-6494.1996.tb00514.x

McAdams, D. P., & McLean, K. C. (2013). Narrative identity. *Current Directions in Psychological Science, 22,* 233–238. http://dx.doi.org/10.1177/0963721413475622

McAdams, D. P., & Pals, J. L. (2006). A new Big Five: Fundamental principles for an integrative science of personality. *American Psychologist, 61,* 204–217. http://dx.doi.org/10.1037/0003-066X.61.3.204

McAdams, D. P., Reynolds, J., Lewis, M., Patten, A. H., & Bowman, P. J. (2001). When bad things turn good and good things turn bad: Sequences of redemption and contamination in life narrative and their relation to psychosocial adaptation in midlife adults and in students. *Personality and Social Psychology Bulletin, 27,* 474–485. http://dx.doi.org/10.1177/0146167201274008

McLean, K. C., Pasupathi, M., & Pals, J. L. (2007). Selves creating stories creating selves: A process model of self-development. *Personality and Social Psychology Review, 11,* 262–278. http://dx.doi.org/10.1177/1088868307301034

Nezlek, J. B. (2011). *Multilevel modeling for social and personality psychology: The Sage library in social and personality psychology methods*. London: Sage Publications.

Oltmanns, T. F., Rodrigues, M. M., Weinstein, Y., & Gleason, M. E. J. (2014). Prevalence of personality disorders at midlife in a community sample: Disorders and symptoms reflected in interview, self, and informant reports. *Journal of Psychopathology and Behavioral Assessment, 36,* 177–188. http://dx.doi.org/10.1007/s10862-013-9389-7

Pals, J. L. (2006). Narrative identity processing of difficult life experiences: Pathways of personality development and positive self-transformation in adulthood. *Journal of Personality, 74,* 1079–1110. http://dx.doi.org/10.1111/j.1467-6494.2006.00403.x

Pasupathi, M. (2001). The social construction of the personal past and its implications for adult development. *Psychological Bulletin, 127,* 651–672. http://dx.doi.org/10.1037/0033-2909.127.5.651

Pennebaker, J. W. (1997). Writing about emotional experiences as a therapeutic process. *Psychological Science, 8,* 162–166. http://dx.doi.org/10.1111/j.1467-9280.1997.tb00403.x

Pennebaker, J. W., & Chung, C. K. (2011). Expressive writing and its links to mental and physical health. In H. S. Friedman (Ed.), *Oxford handbook of health psychology* (pp. 417–437). New York, NY: Oxford University Press.

Petrie, K. J., & Weinman, J. (2012). Patients' perceptions of their illnesses: The dynamo of volition in health care. *Current Directions in Psychological Science, 21,* 60–65. http://dx.doi.org/10.1177/0963721411429456

Philippe, F. L., Koestner, R., Beaulieu-Pelletier, G., & Lecours, S. (2011). The role of need satisfaction as a distinct and basic psychological component of autobiographical memories: A look at well-being. *Journal of Personality, 79,* 905–938. http://dx.doi.org/10.1111/j.1467-6494.2010.00710.x

Ryff, C. D., & Almeida, D. M. (2009). *National Survey of Midlife in the United States (MIDUS II): Daily Stress Project, 2004–2009: Scales*. Ann Arbor, MI: Inter-University Consortium for Political and Social Research.

Ryff, C. D., & Keyes, C. L. M. (1995). The structure of psychological well-being revisited. *Journal of Personality and Social Psychology, 69,* 719–727. http://dx.doi.org/10.1037/0022-3514.69.4.719

Sarbin, T. (1986). *Narrative psychology: The storied nature of human conduct*. New York, NY: Praeger.

Shrout, P. E., & Fleiss, J. L. (1979). Intraclass correlations: Uses in assessing rater reliability. *Psychological Bulletin, 86,* 420–428. http://dx.doi.org/10.1037/0033-2909.86.2.420

Simon, G. E., Revicki, D. A., Grothaus, L., & Vonkorff, M. (1998). SF-36 summary scores: Are physical and mental health truly distinct? *Medical Care, 36,* 567–572. http://dx.doi.org/10.1097/00005650-199804000-00012

Singer, J. A. (2004). Narrative identity and meaning making across the adult lifespan: An introduction. *Journal of Personality, 72,* 437–460. http://dx.doi.org/10.1111/j.0022-3506.2004.00268.x

Singer, J. D., & Willett, J. B. (2003). *Applied longitudinal data analysis: Modeling change and event occurrence*. New York, NY: Oxford University Press. http://dx.doi.org/10.1093/acprof:oso/9780195152968.001.0001

Tavernier, R., & Willoughby, T. (2012). Adolescent turning points: The association between meaning-making and psychological well-being. *Developmental Psychology, 48,* 1058–1068. http://dx.doi.org/10.1037/a0026326

Thorne, A., & Michaelieu, Q. (1996). Situating adolescent gender and self-esteem with personal memories. *Child Development, 67,* 1374–1390. http://dx.doi.org/10.2307/1131706

Ware, J. E., Kosinski, M., & Keller, S. D. (1994). *SF-36 Physical and mental health summary scales: A user's manual*. Boston, MA: The Health Institute, New England Medical Center.

Webster, N. C. (2005, June 13). Color-coded causes. *Advertising Age*. Retrieved from http://adage.com/article/feature-cause-marketing/color-coded/103463/

Weissman, M. M., & Bothwell, S. (1976). Assessment of social adjustment by patient self-report. *Archives of General Psychiatry, 33,* 1111–1115. http://dx.doi.org/10.1001/archpsyc.1976.01770090101010

Weston, S. J. (2014). *Redemption and linguistic analysis: A comparison of methods to understand life stories*. Manuscript submitted for publication.

Woike, B., & Polo, M. (2001). Motive-related memories: Content, structure, and affect. *Journal of Personality, 69,* 391–415. http://dx.doi.org/10.1111/1467-6494.00150

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## Correction to Masuda et al. (2008)

In the article, "Placing the Face in Context: Cultural Differences in the Perception of Facial Emotion," by Takahiko Masuda, Phoebe C. Ellsworth, Batja Mesquita, Janxin Leu, Shigehito Tanida, and Ellen Van de Veerdonk (*Journal of Personality and Social Psychology*, 2008, Vol. 94, No. 3, pp. 365–381. http://dx.doi.org/10.1037/0022-3514.94.3.365), various corrections are needed for several subsections in Study 2 and Figure 3.

On page 374 in the original article, the subsection title *Effects of Others' Emotions on Judgments of the Central Person's Emotion* should appear instead as *Perception of the Central Person's Emotions*. On page 375, a word in the first sentence in the *Anger* subsection in Study 2 was omitted and should appear instead as "The hypothesis that Japanese emotion judgments of anger would reflect the emotions of the background figures, whereas American emotion judgments would not was marginally borne out." In Line 9 in the *Anger* subsection, the parenthetical content should read "(e.g., an angry target and angry backgrounds)." In Line 10 in the *Anger* subsection, the parenthetical content should read "(e.g., an angry target and happy backgrounds)." Line 11 in the *Anger* subsection should read instead as ". . . was marginally larger than that of Americans, . . ." In Line 4 in the *Sadness* subsection, the 4 (background) should appear instead as 3 (background). The parenthetical content in Line 9 in the *Sadness* subsection should read instead as "(a sad target and sad backgrounds)." The parenthetical content in Line 10 in the *Sadness* subsection should read instead as "(e.g., a sad target and happy backgrounds)." In Line 5 in the *Happiness* subsection, 4 (background) should appear instead as 3 (background). The degree of freedom for the *F* test in Line 7 in the *Happiness* subsection should appear instead as "$F(2, 47)$."

On page 376, the colors of the bar graph in Figure 3 are incorrect. All dark gray bars should be white, and all white bars should be gray.

The online version of this article has been corrected.

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